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Dipartimento di Informatica e Automazione
Computer Networks Research Group

netkit lab

bgp: transit as

Version	1.7.1 (compact)
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Description	possible architectures for a transit provider, bad interactions between igp and bgp routing protocols, configuration of tunnels

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disclaimer

- this is a rather complex lab
- observed phenomena may be due to zebra-specific implementation choices and may not apply to more recent releases of zebra and/or to physical routers
- please read carefully through this documentation before concluding that the lab is not working properly

scenario



- a transit as
  - receives and propagates the full bgp routing table from/to its neighbors (customers, peers, providers)
  - receives and forwards traffic across its neighbors

transit as: degrees of freedom

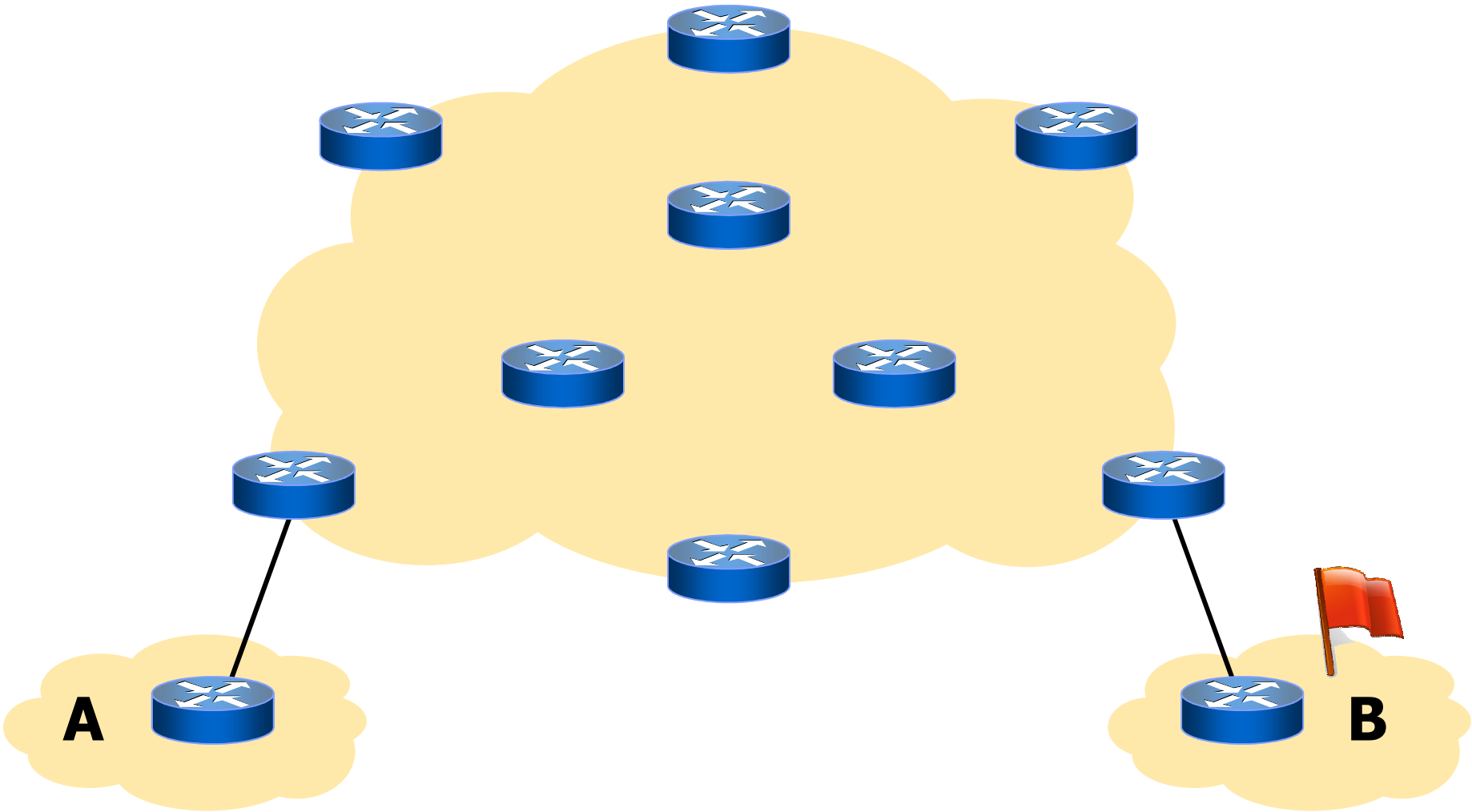
- internal routers must support traffic flows from/to neighboring ases
  - choice 1: redistribute bgp routes into the igp
    - overgrowth of igp routing tables 
    - update churn from bgp affects the igp 
  - choice 2: route traffic flowing through via an ad-hoc overlay
    - internal routers know about border routers only

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choice 1

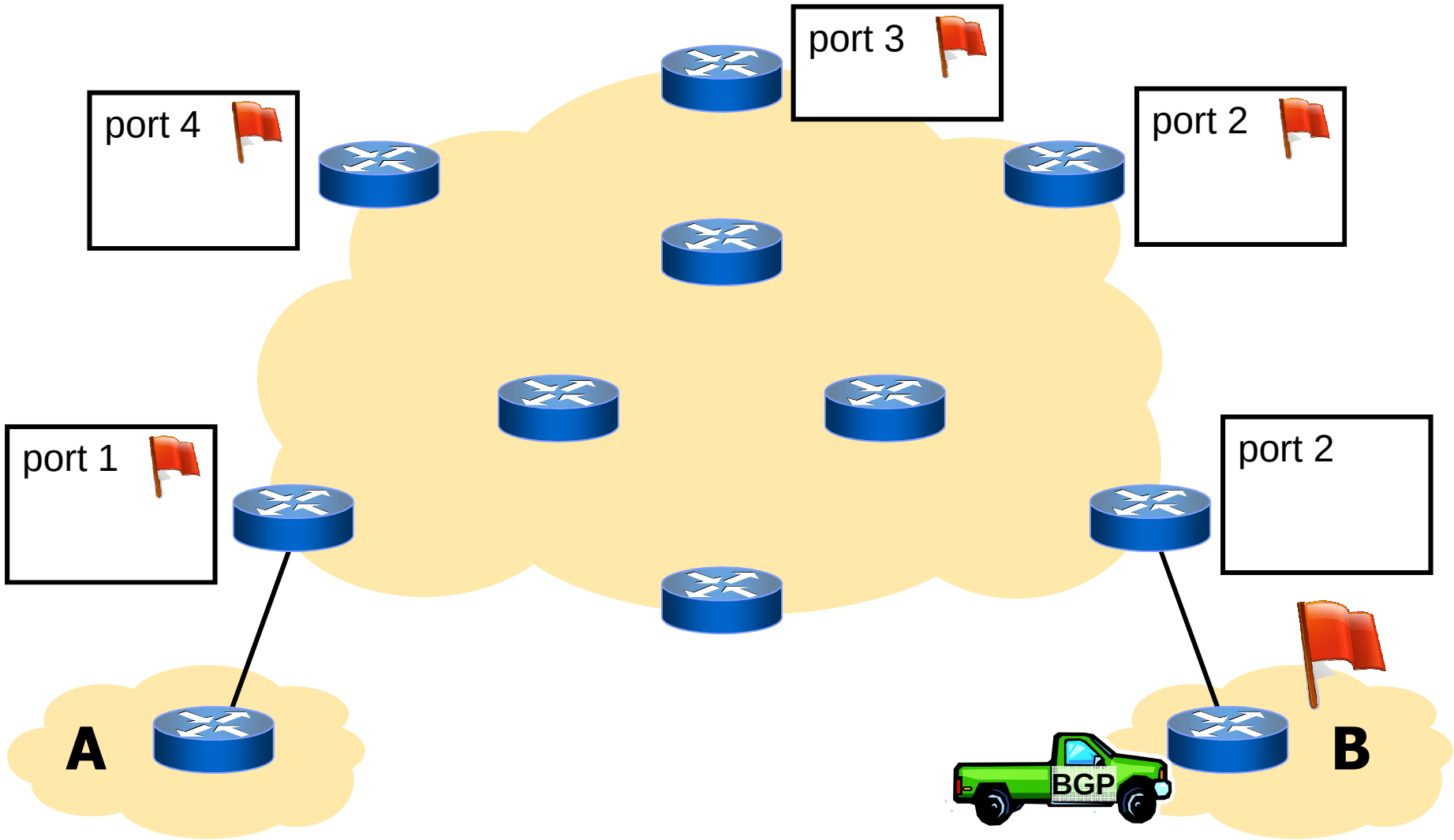
redistribution

a reason to redistribute



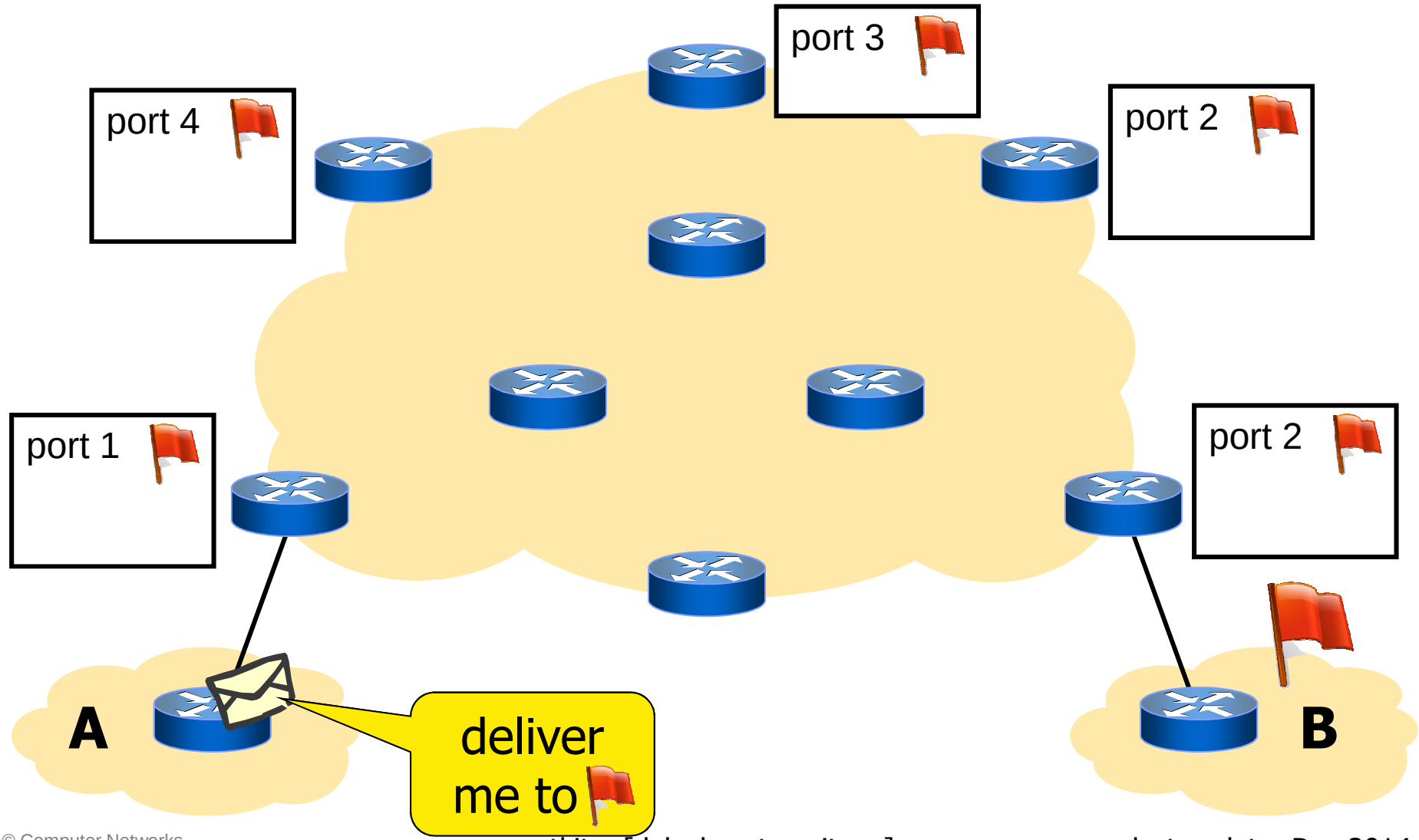
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a reason to redistribute



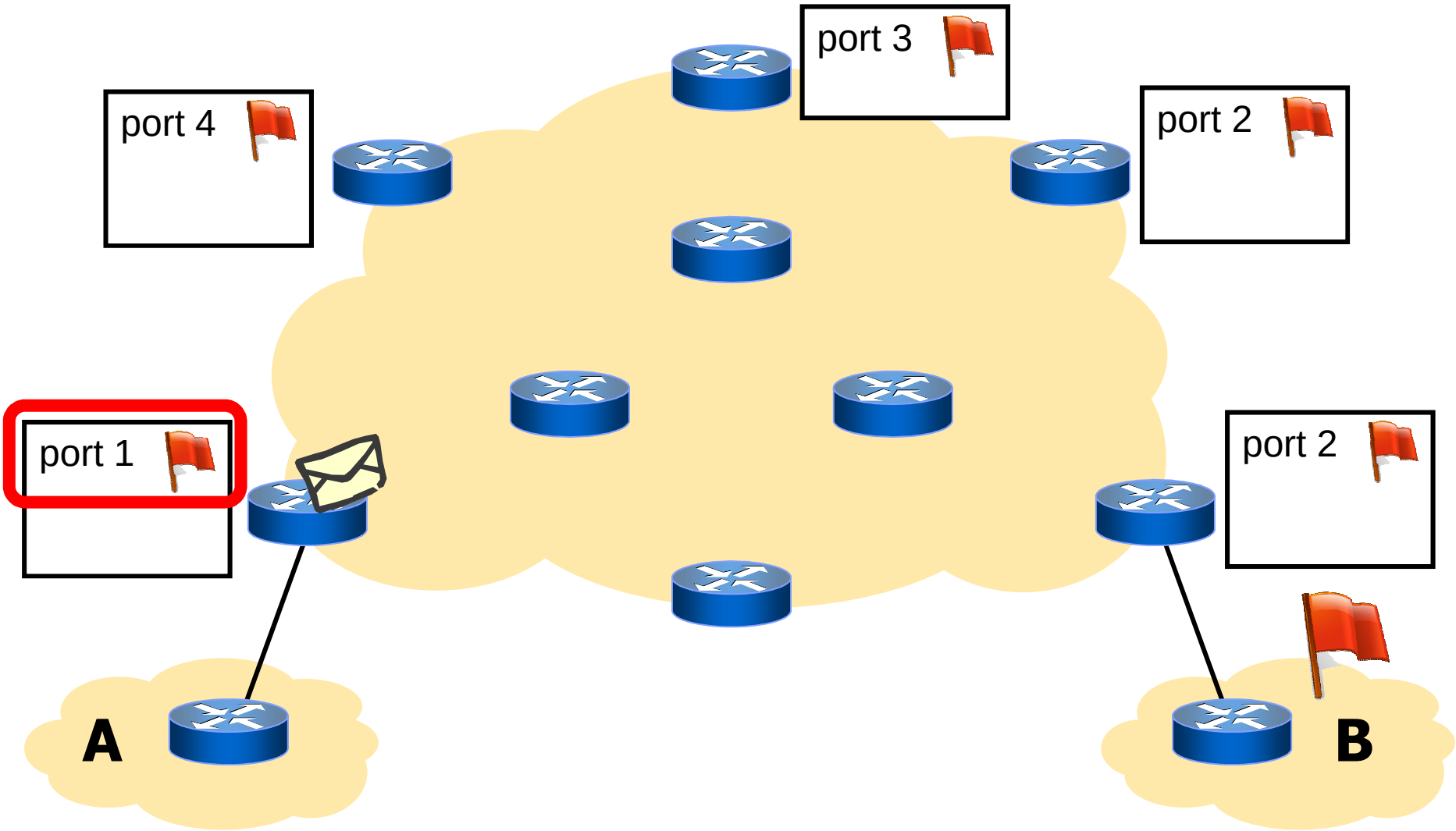
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a reason to redistribute



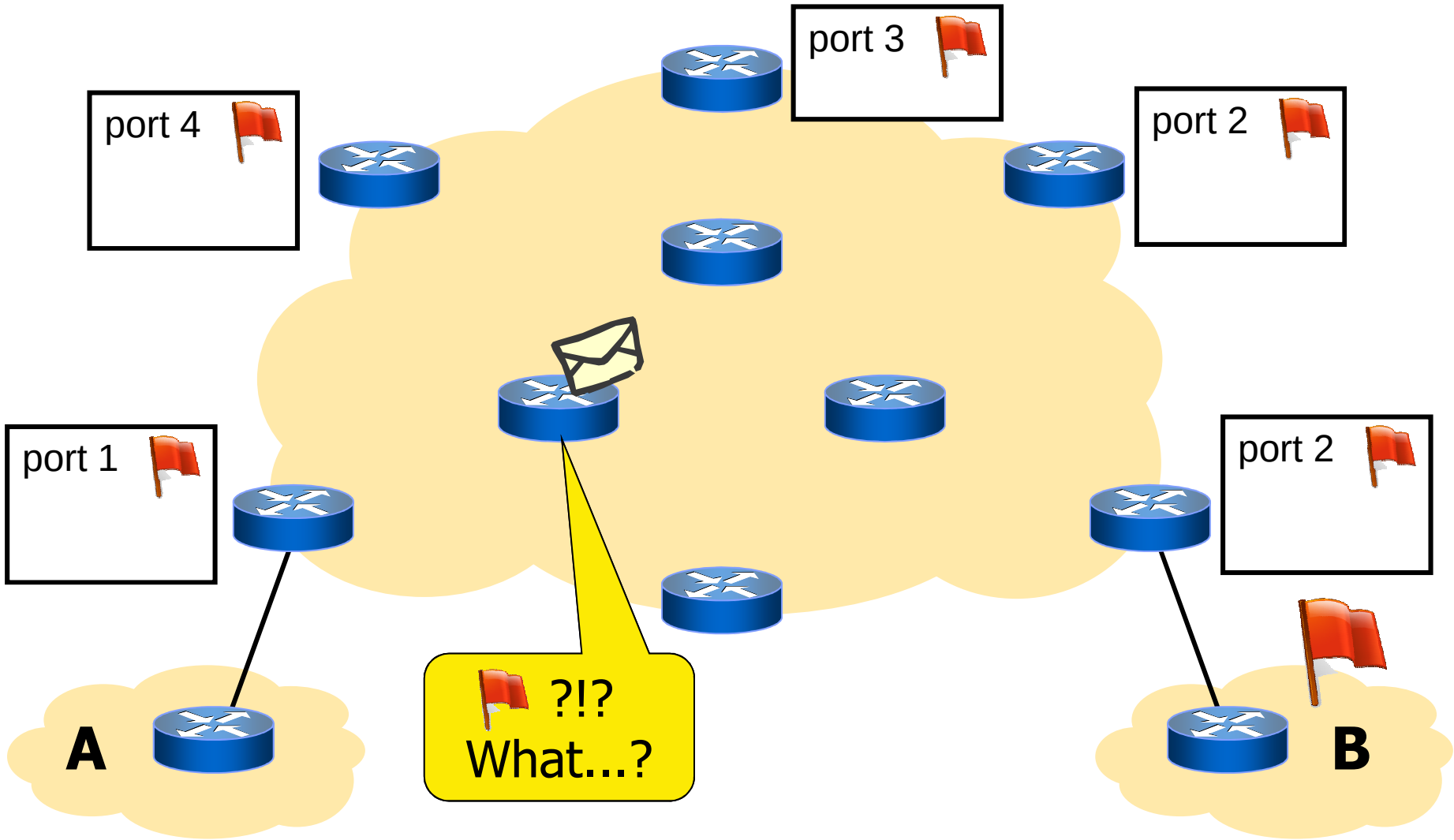
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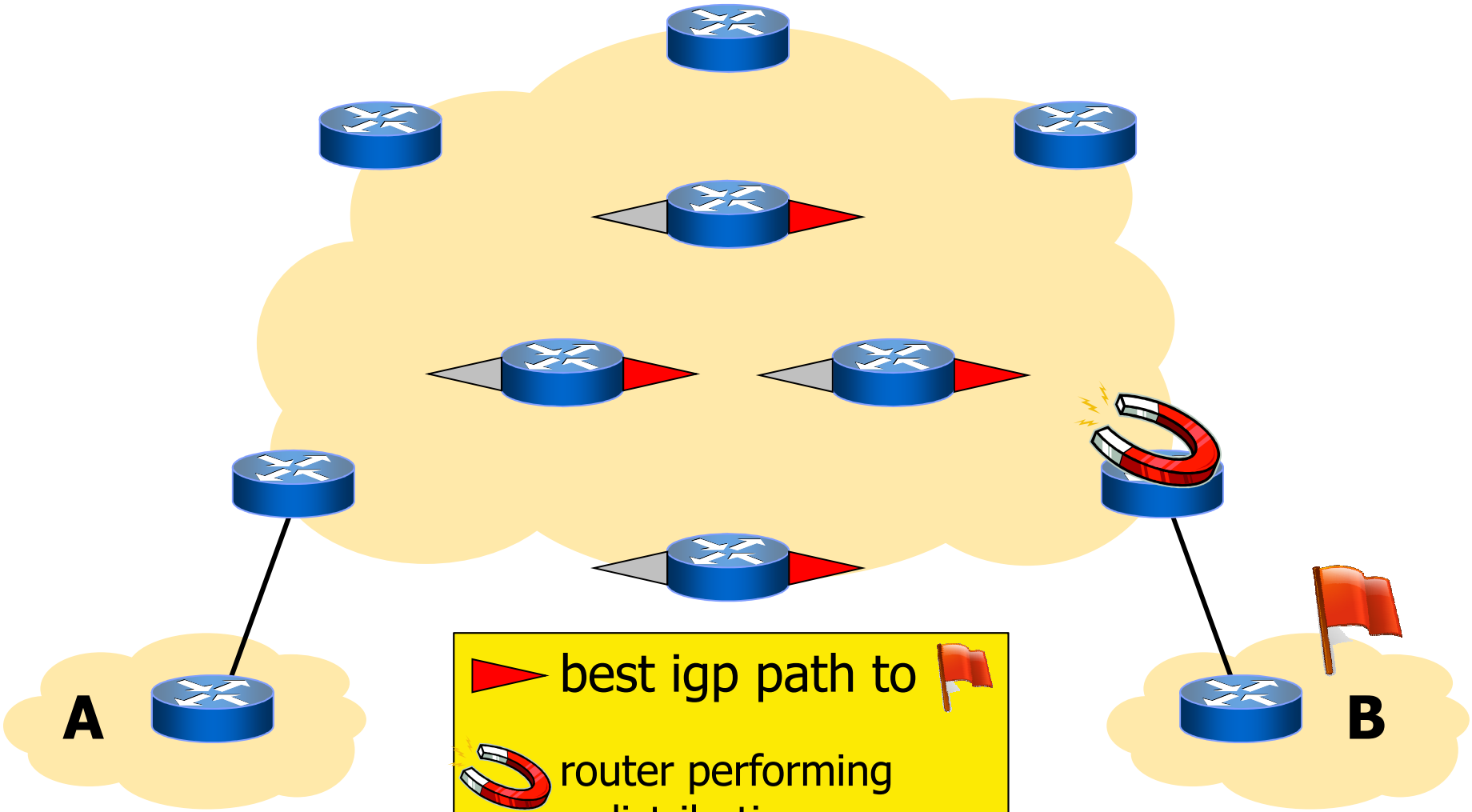


how to redistribute
bgp → rip

zebra rip configuration file

```
router rip
   network eth1
   redistribute connected
   redistribute bgp
```

beware of redistributing ibgp!

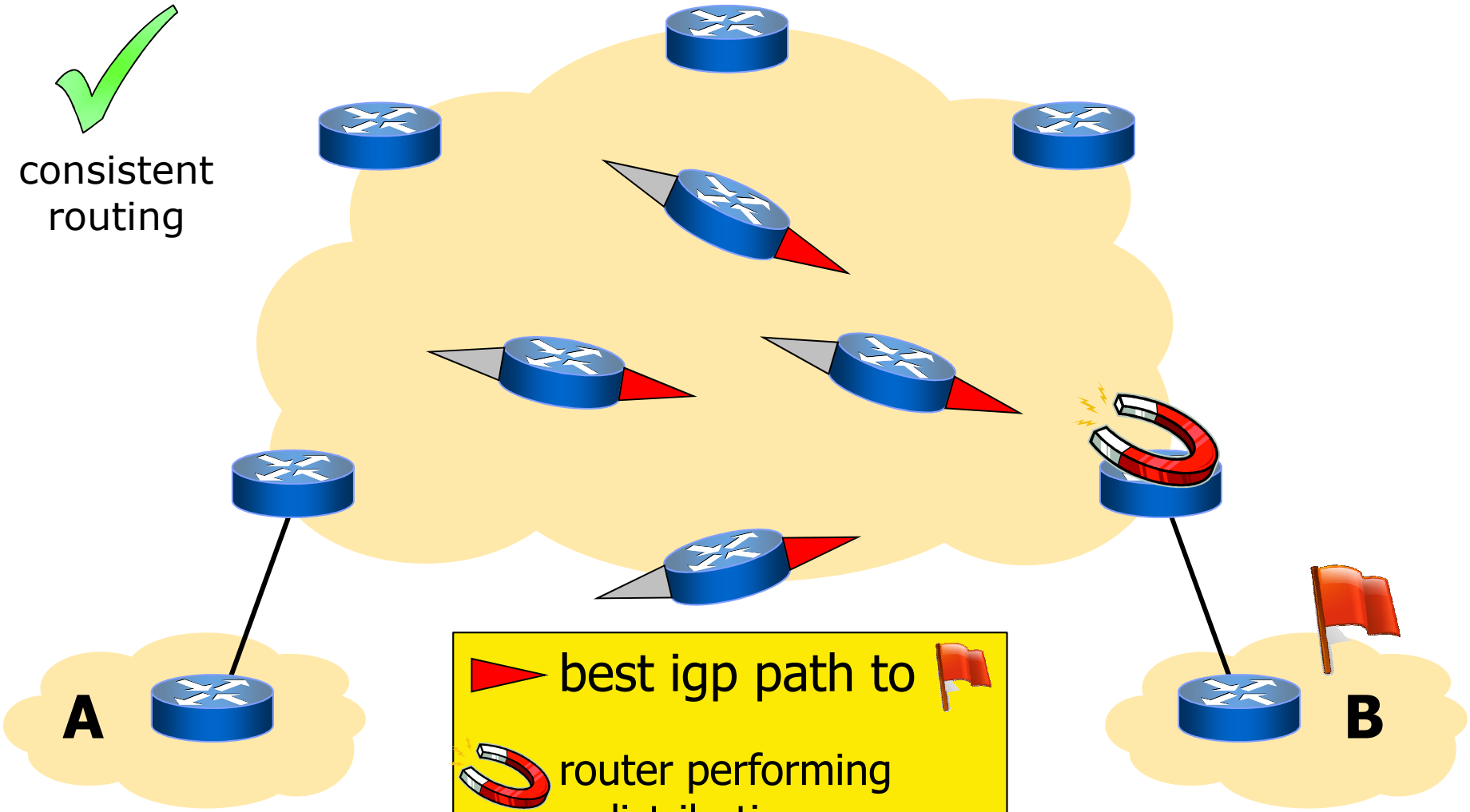


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beware of redistributing ibgp!

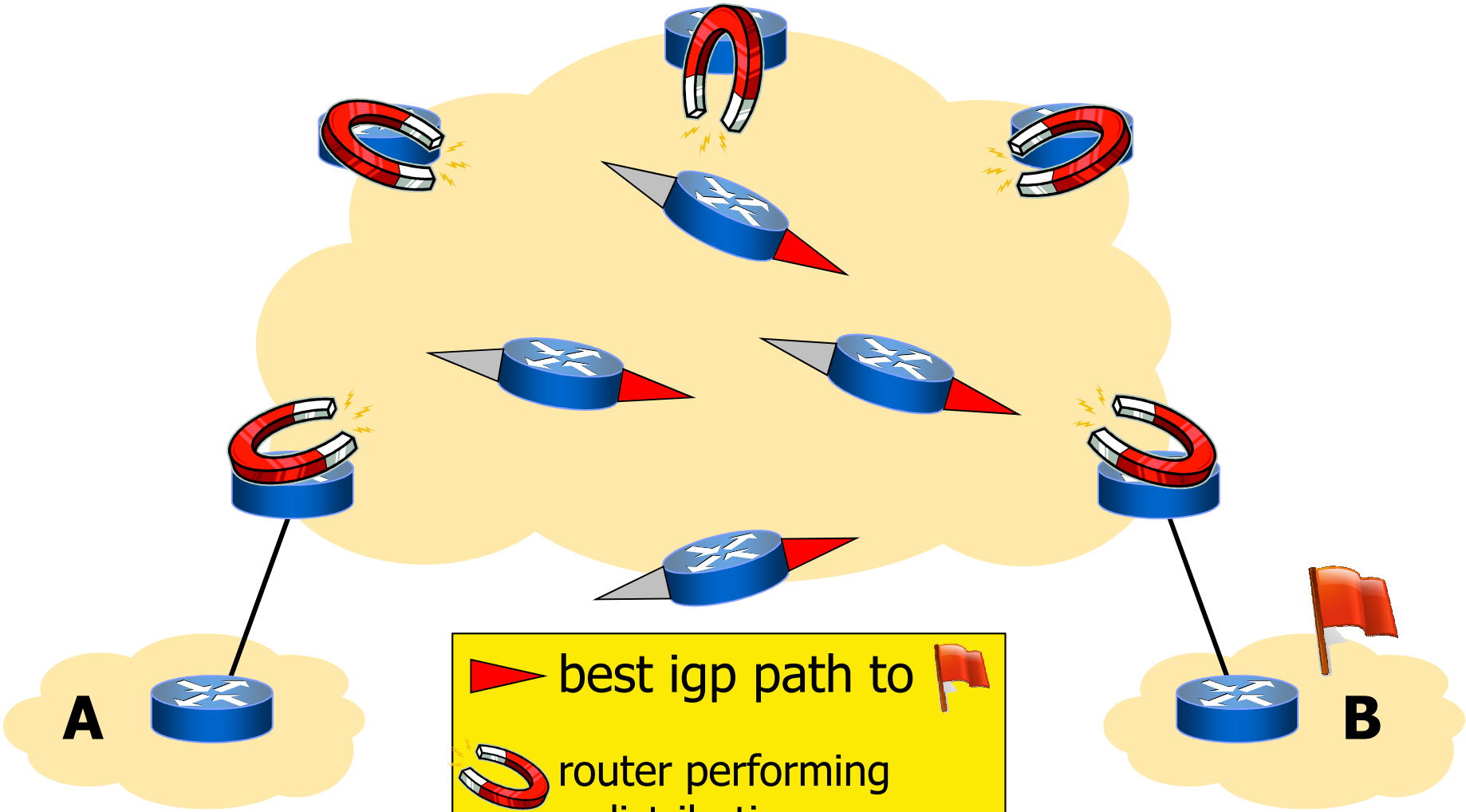
consistent routing



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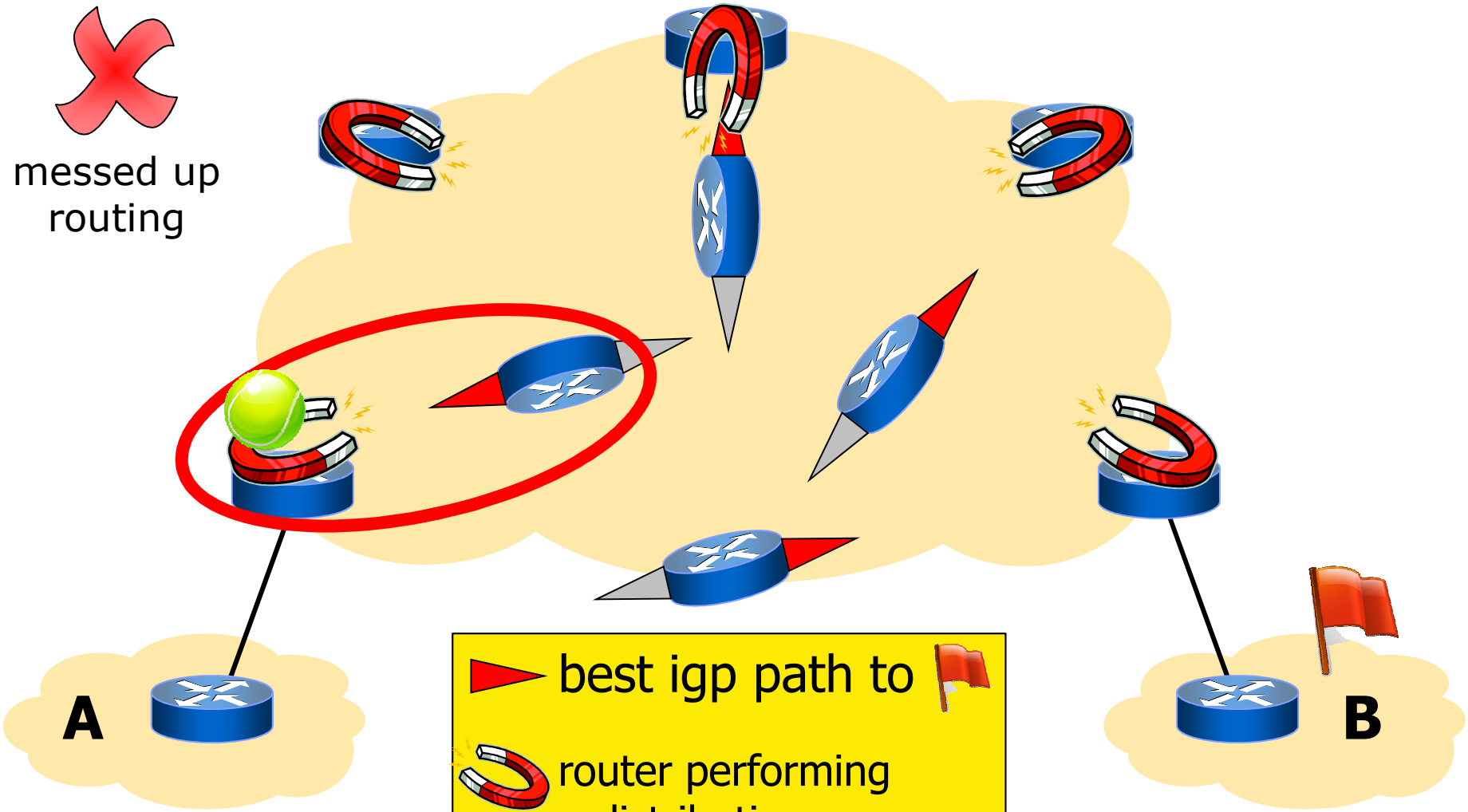
beware of redistributing ibgp!



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beware of redistributing ibgp!



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beware of redistributing ibgp!

- cisco (and juniper) say:

By default, iBGP redistribution into IGP is disabled. To enable redistribution of iBGP routes into IGP, issue the **`bgp redistribute-internal`** *command. Precautions should be taken to redistribute specific routes using route maps into IGP. Note: Redistributing internal Border Gateway Protocol (iBGP) routes into an Interior Gateway Protocol may cause routing loops within the Autonomous System (AS). This is not recommended. Route filters should be set to control the information which is imported into the IGP.*



https://supportforums.cisco.com/document/8621/unable-redistribute-ibgp-learnt-routes-igp-such-eigrp-ospf-and

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how to redistribute ebgp → rip

- no way in zebra to say "redistribute ebgp" but...
- ...**route-maps** can be applied on redistributed routes...
  - ...for example based on recognizing ebgp next-hops

zebra rip configuration file

```
ip prefix-list myNeighbors permit 12.0.0.0/30 le 32
route-map eBGP permit 10
    match ip next-hop prefix-list myNeighbors
router rip
    network eth1
    redistribute connected
    redistribute bgp route-map eBGP
```

how to redistribute **e**bgp → rip

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```

how to redistribute ebgp → rip

- no way in zebra to say "redistribute ebgp" but...
- ...**route-maps** ted routes...
  - ...for example b ops

this is the BGP next-hop (i.e., *before* the recursive lookup)

zebra rip configuration file

```
ip prefix-list myNeighbors permit 12.0.0.0/30 le 32
route-map eBGP permit 10
   match ip next-hop prefix-list myNeighbors
router rip
   network eth1
   redistribute connected
   redistribute bgp route-map eBGP
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how to redistribute **e**bgp → rip

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```

how to redistribute **e**bgp → rip

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- ...**route-maps** can be applied on redistributed routes...
  - ...for example based on recognizing ebgp next-hops

zebra rip configuration file

```
ip prefix-list myNeighbors permit 12.0.0.0/30 le 32
route-map eBGP permit 10
   match ip next-hop prefix-list myNeighbor
router rip
   network eth1
   redistribute connected
   redistribute bgp route
```

match all the more specifics of the **12.0.0.0/30** network (next-hops are single ip addresses)

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beware of redistributing *bgp



as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network          Next Hop        Metric From            Time
R(n) 1.1.1.1/32       10.0.0.10            3 10.0.0.10      02:58
R(n) 2.2.2.2/32       10.0.0.5             3 10.0.0.5       02:43
R(n) 3.3.3.3/32       11.0.0.10            2 11.0.0.10      02:55
R(n) 10.0.0.0/30      10.0.0.10            2 10.0.0.10      02:58
C(i) 10.0.0.4/30      0.0.0.0              1 self
C(i) 10.0.0.8/30      0.0.0.0              1 self
R(n) 11.0.0.0/30      10.0.0.10            2
R(n) 11.0.0.4/30      10.0.0.5             2
C(i) 11.0.0.8/30      0.0.0.0              1
R(n) 12.0.0.0/30      10.0.0.5             3
R(n) 12.0.0.4/30      11.0.0.10            2
R(n) 12.0.0.8/30      11.0.0.10            2
R(n) 12.0.0.12/30     11.0.0.10            2 11.0.0.10      02:55
R(n) 20.0.0.0/8       10.0.0.5             3 10.0.0.5       02:43
R(n) 30.0.0.0/8       11.0.0.10            2 11.0.0.10      02:55
R(n) 100.0.0.0/8      11.0.0.10            2 11.0.0.10      02:55
as10r6-ripd>
```

routing tables of the internal routers become unnecessarily large

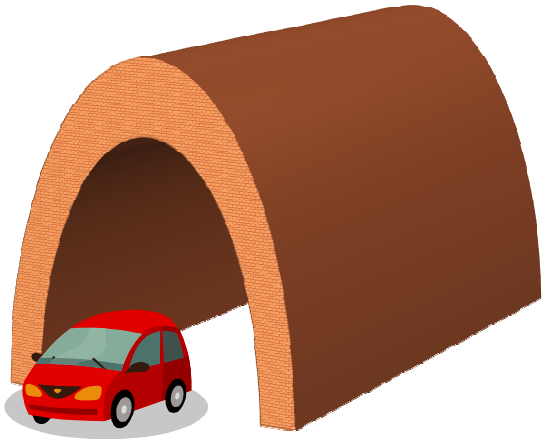
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choice 2

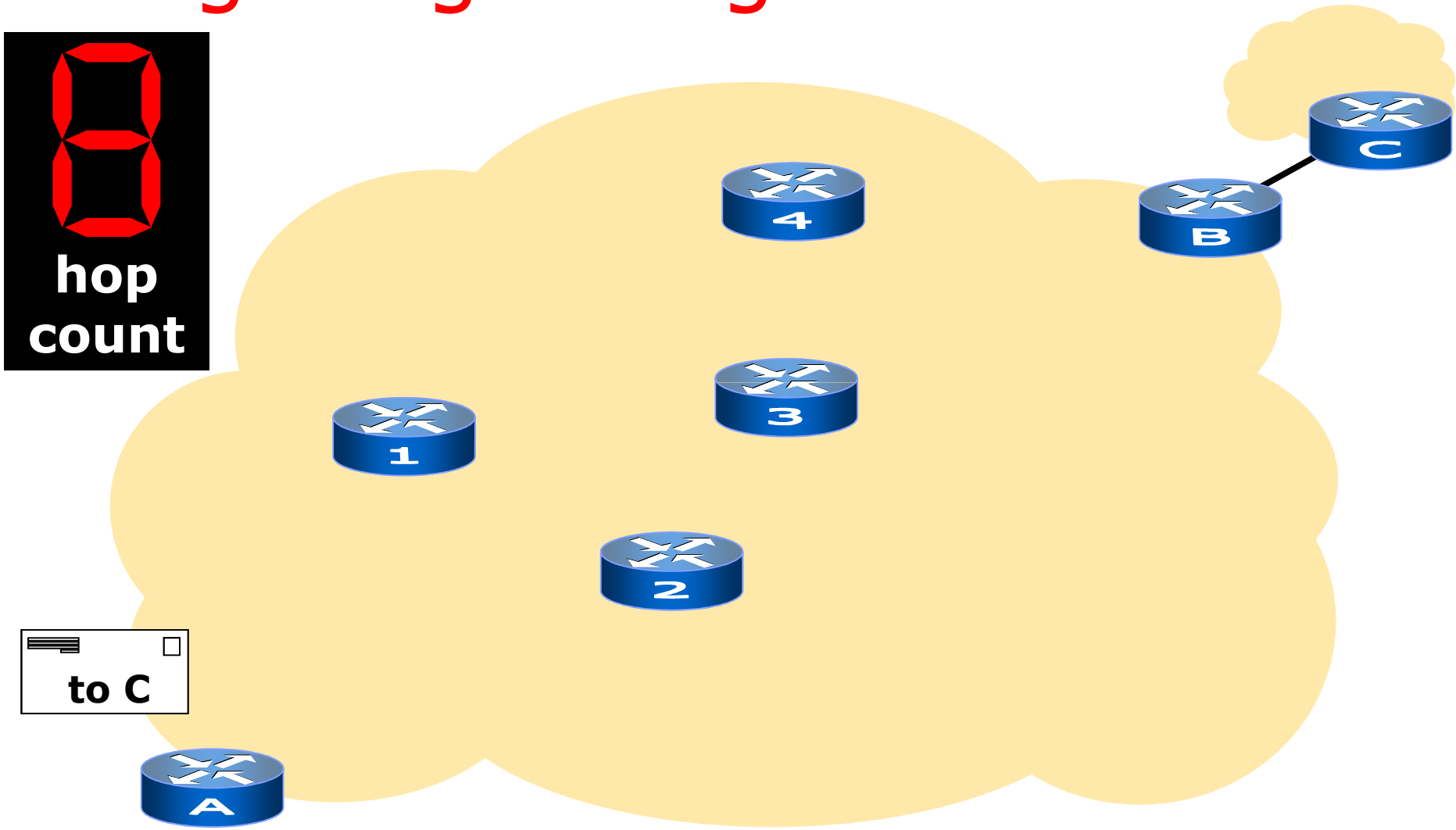
overlay

overlay

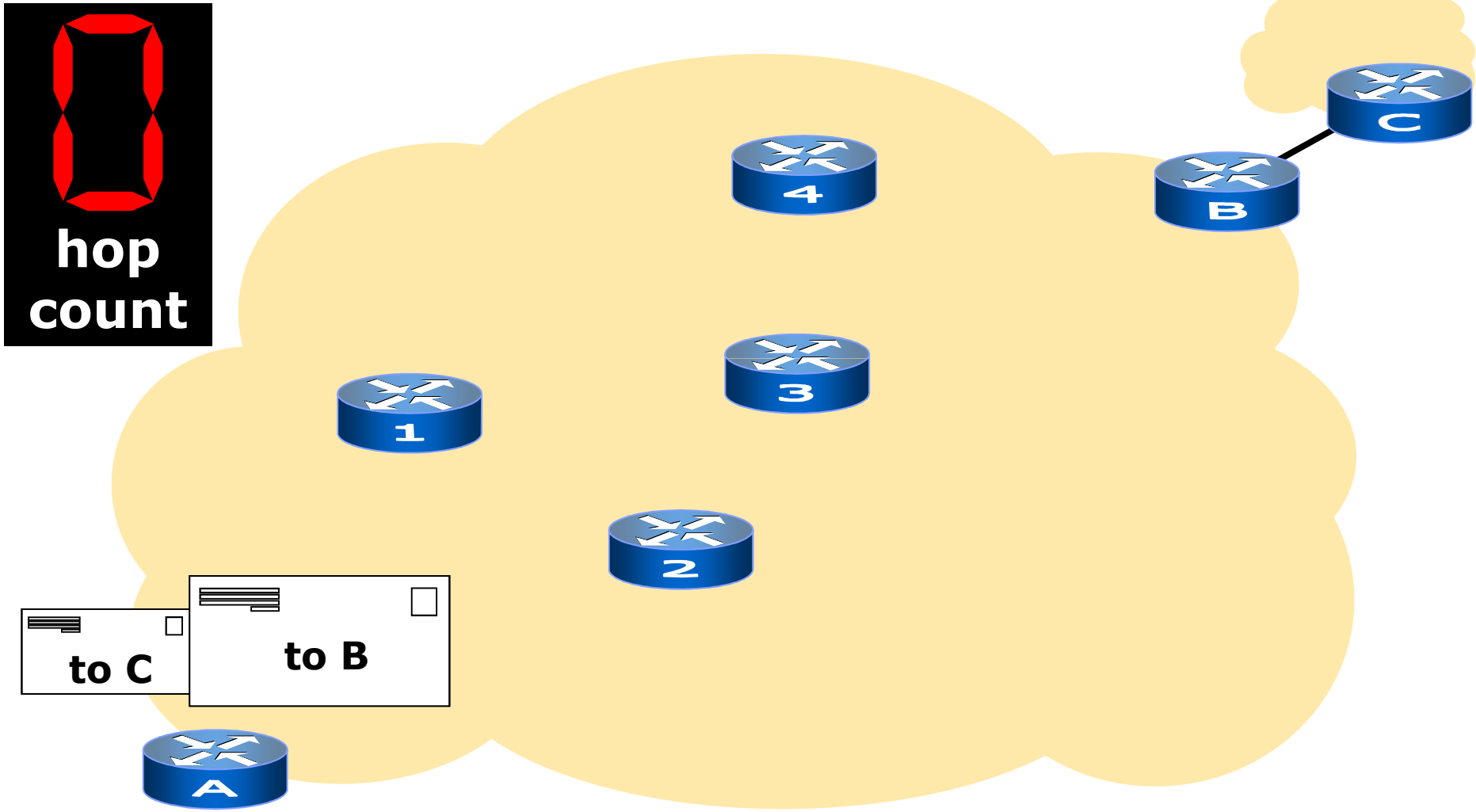
- ebgp is not redistributed into the igp
  - smaller routing tables
  - less igp churn
- ebgp next hops are reached via a direct link (tunnel)



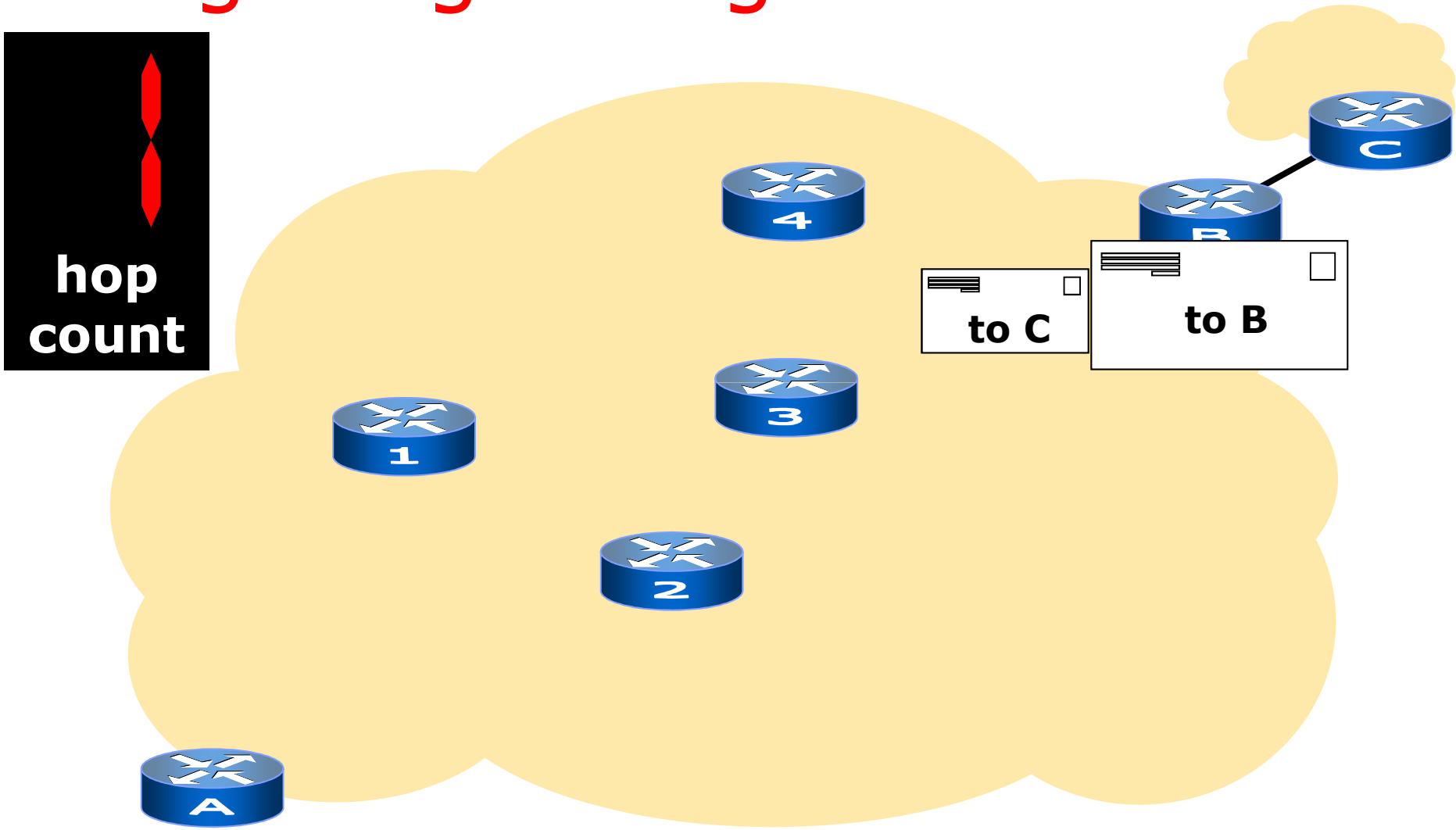
getting through the tunnel



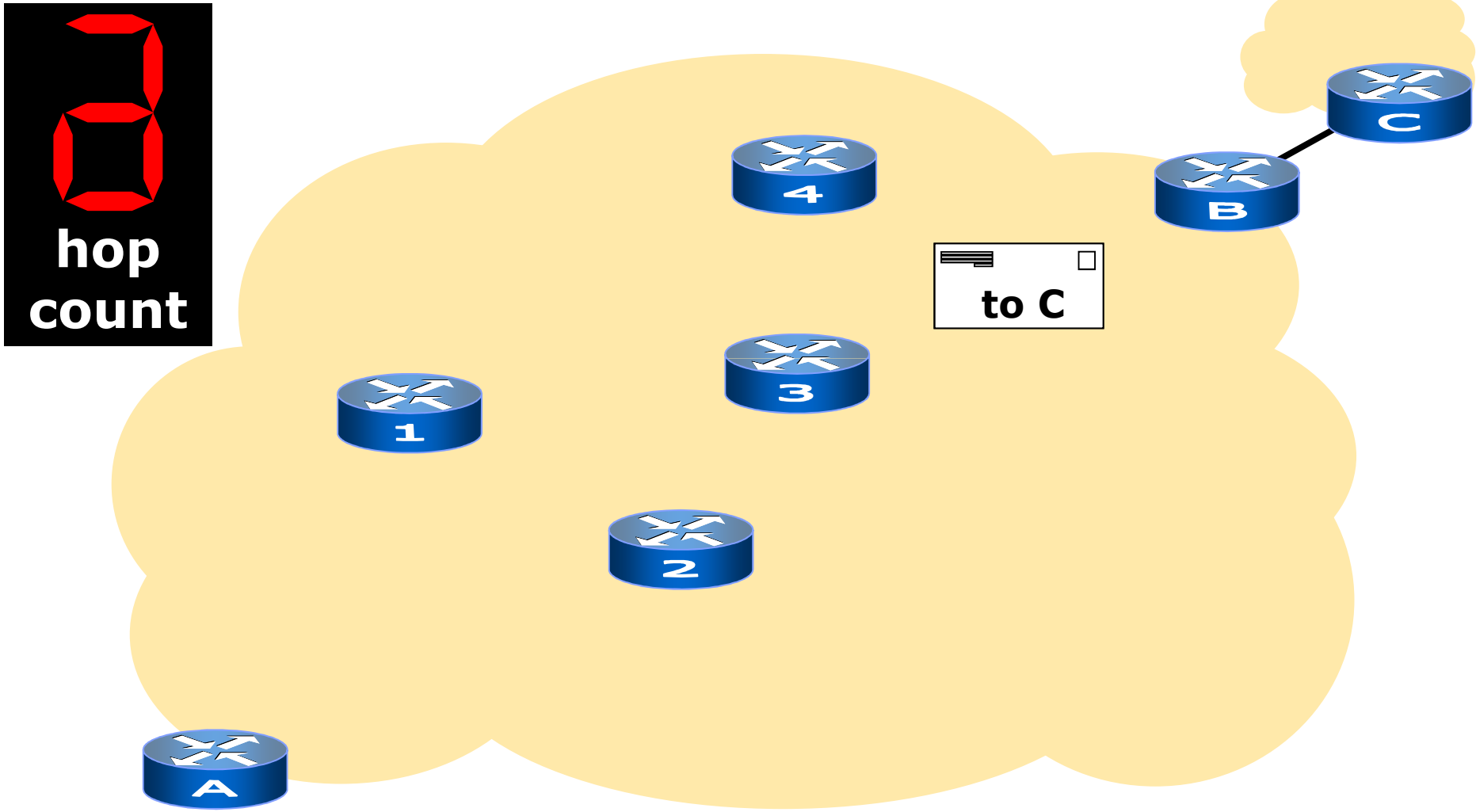
getting through the tunnel



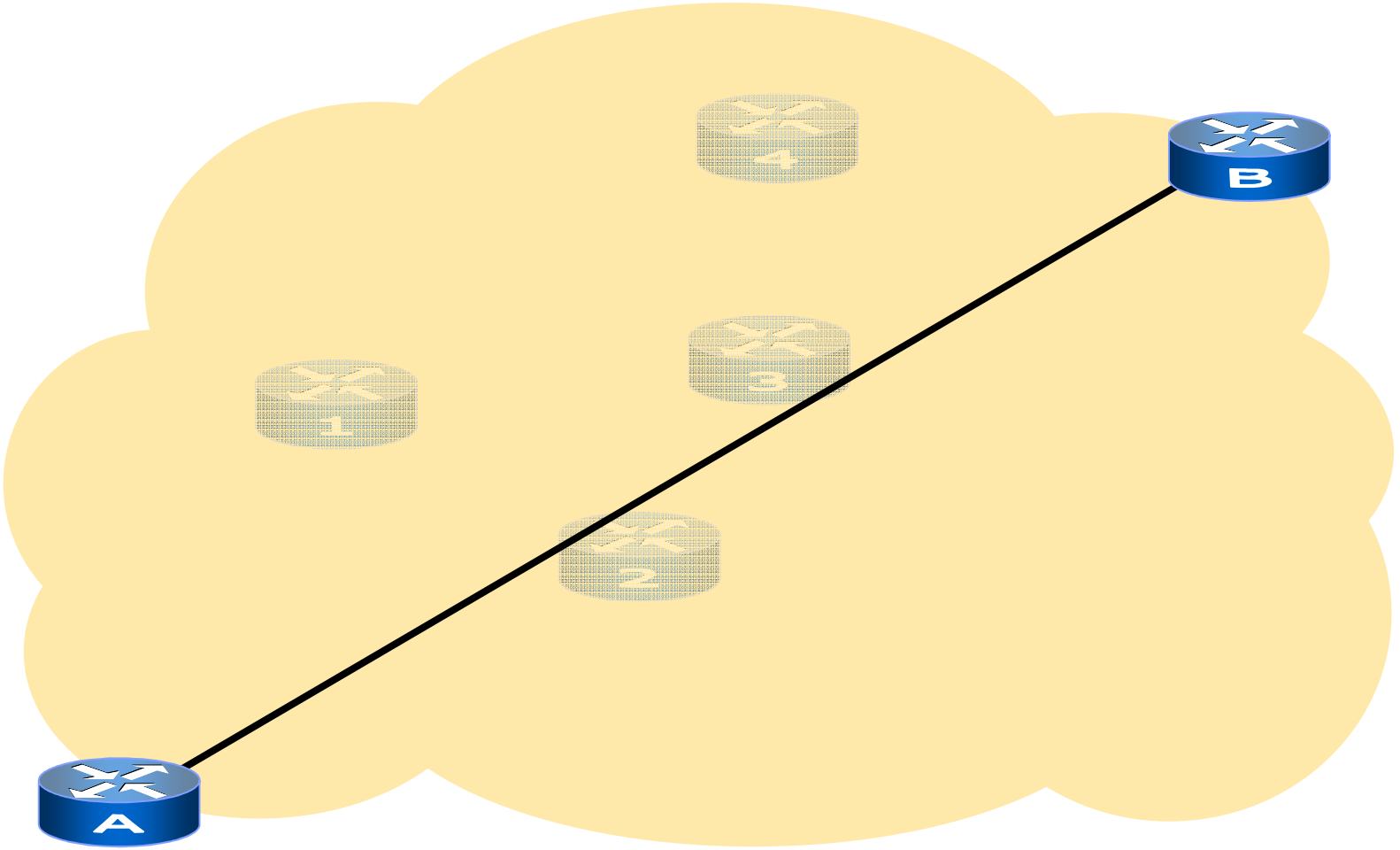
getting through the tunnel



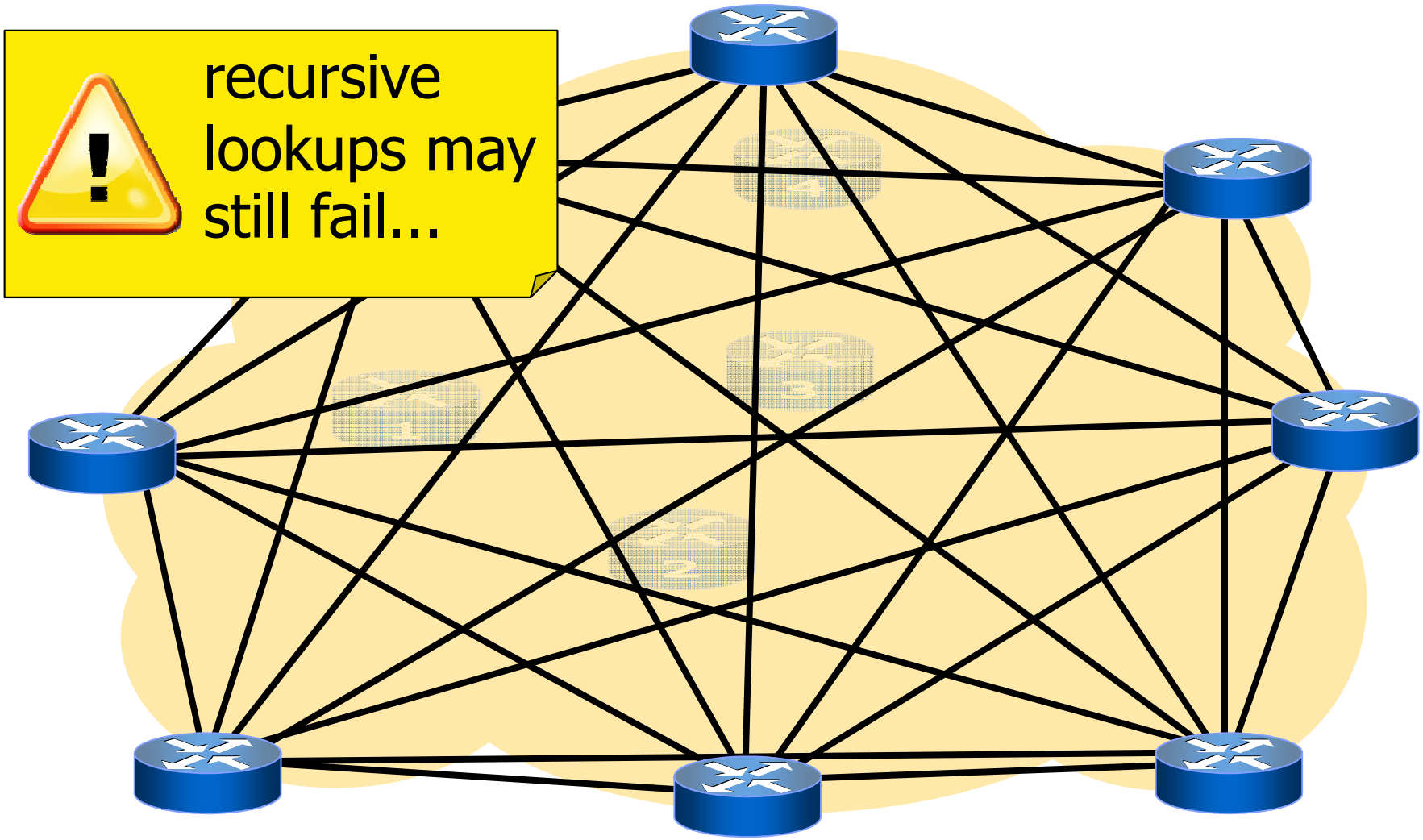
getting through the tunnel



getting through the tunnel



completely isolating bgp from the igp

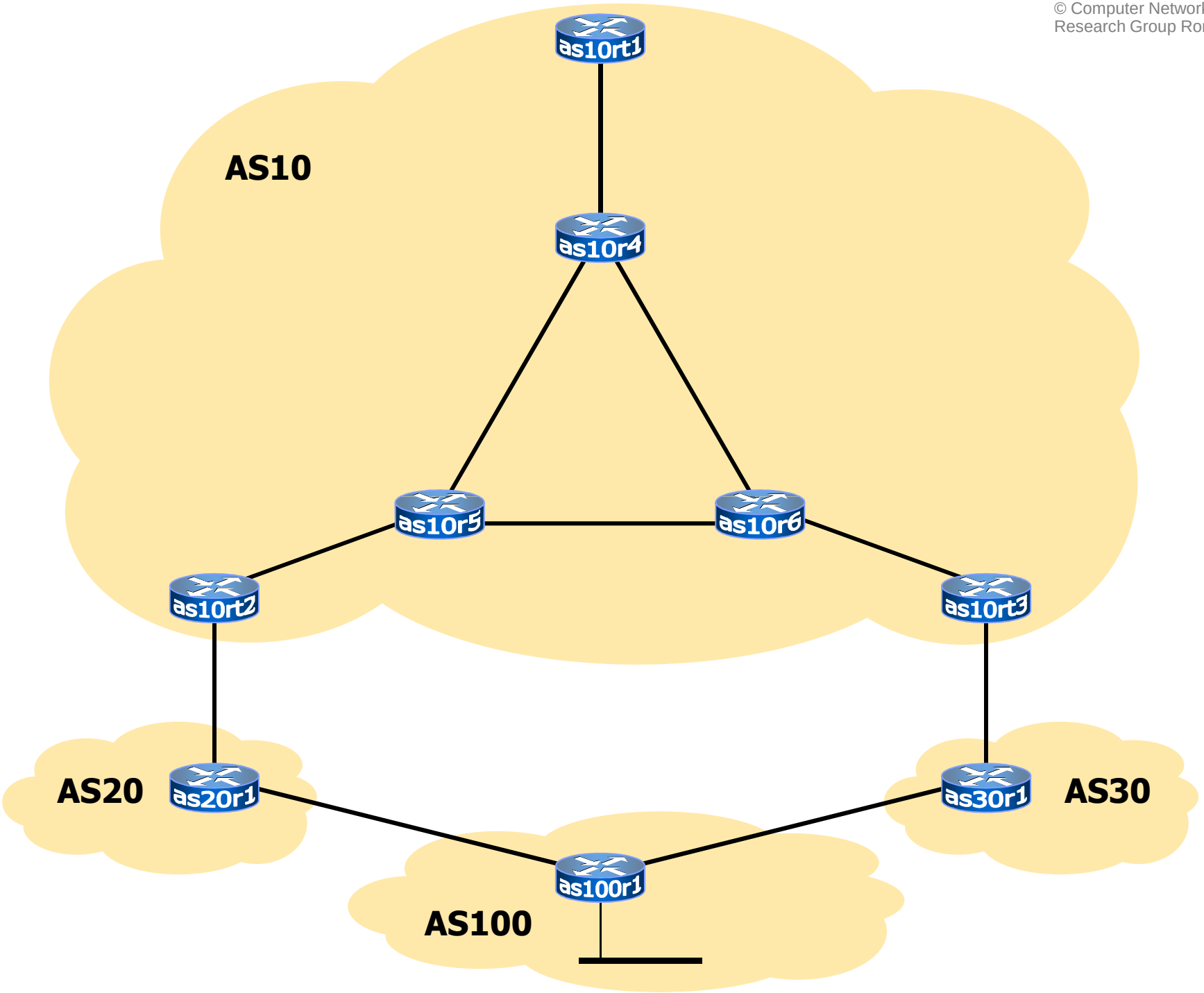


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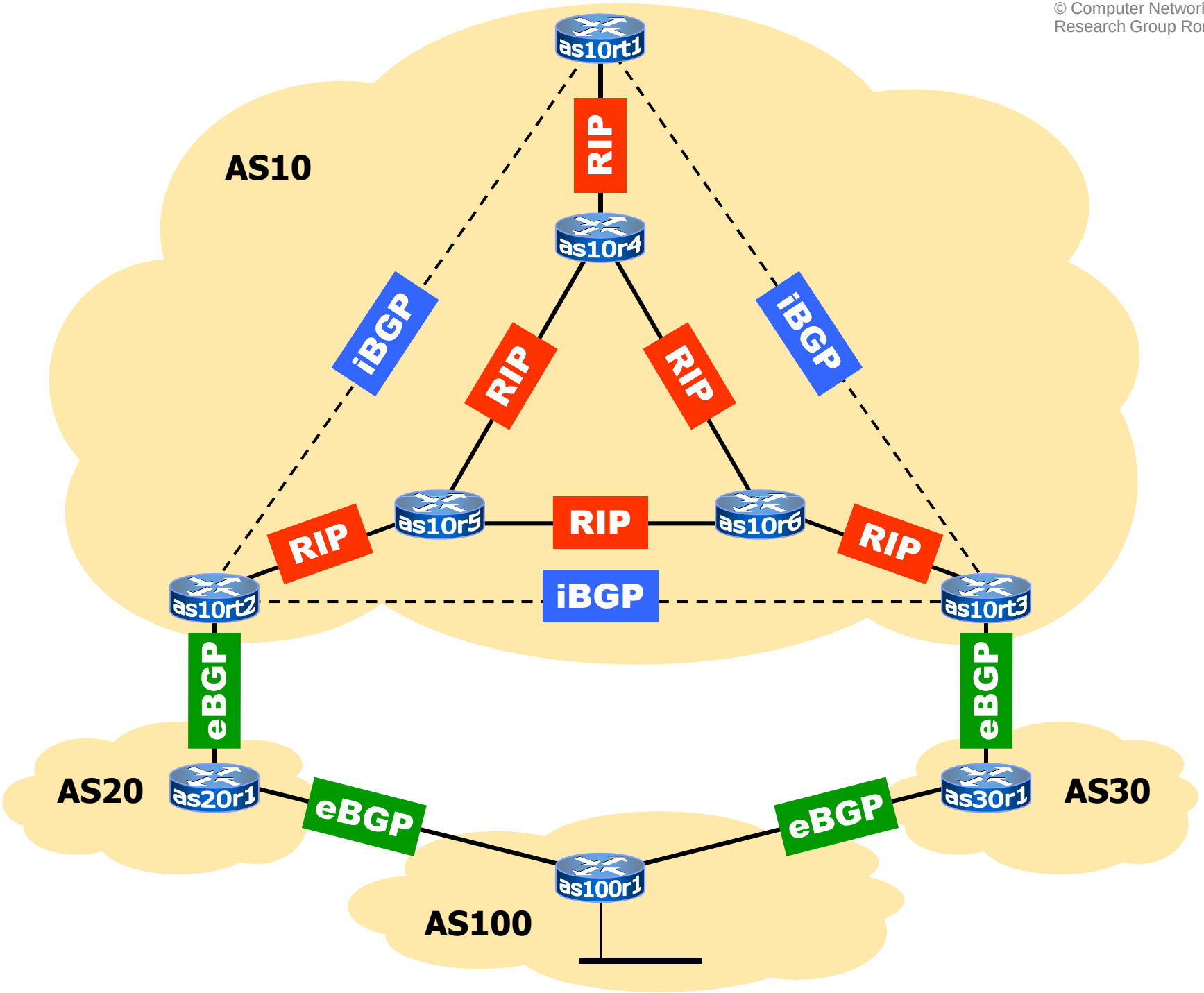
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network topology

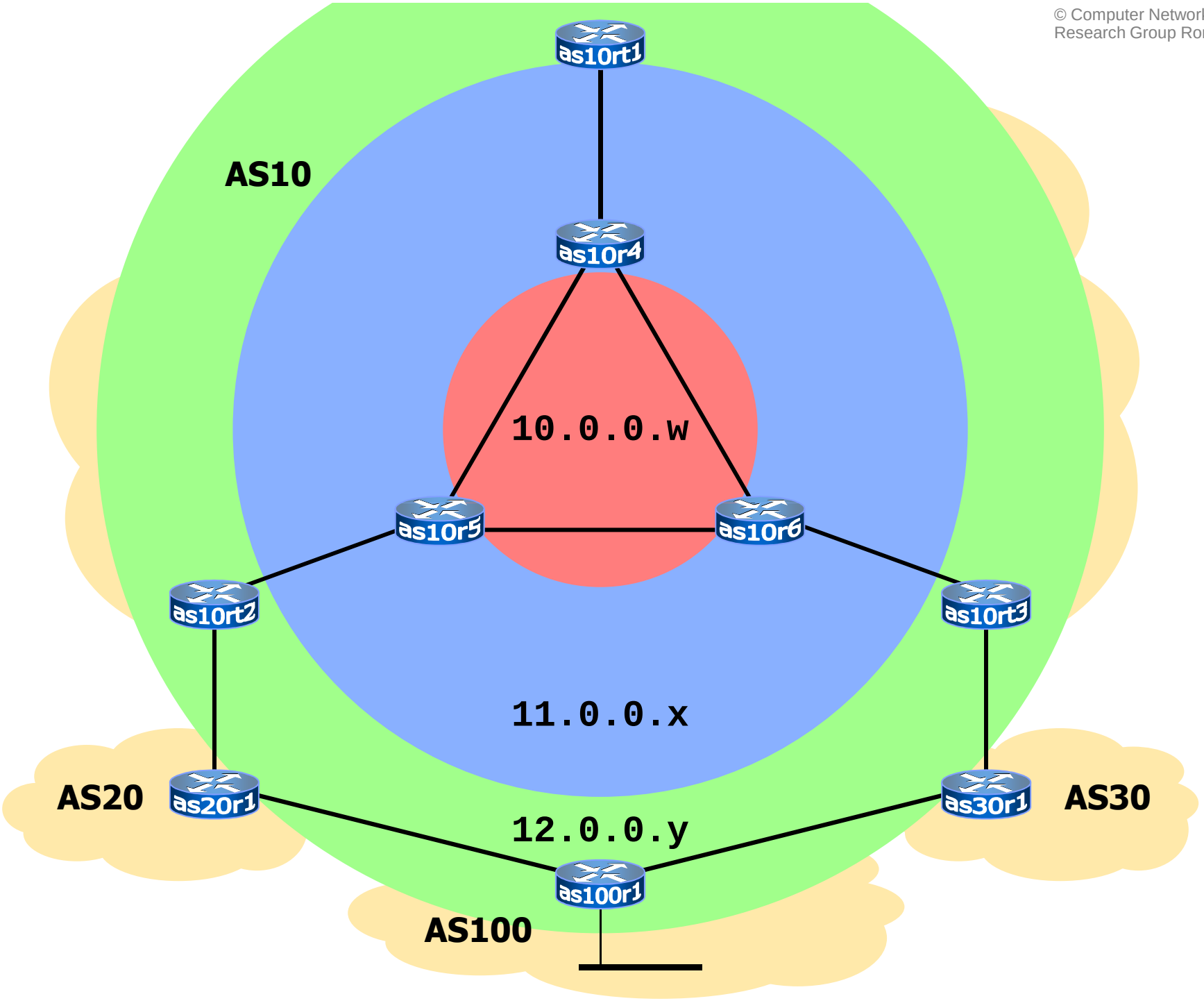
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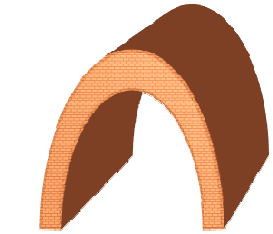
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13.0.0.z

AS10

as10rt1

as10r4

10.0.0.w

as10r5

as10r6

as10rt2

as10rt3

11.0.0.x

AS20

as20r1

as30r1

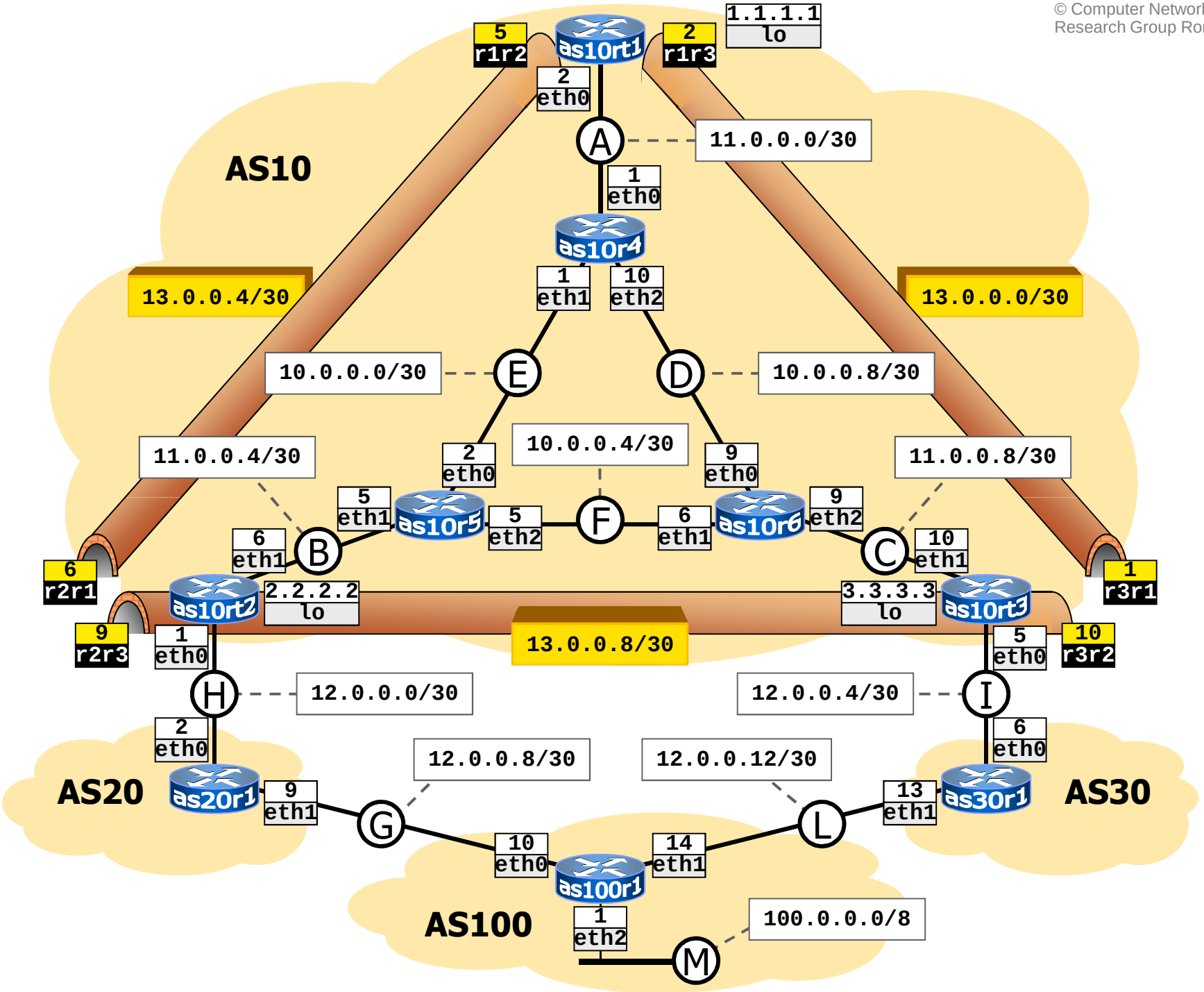
AS30

12.0.0.y

as100r1

AS100

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transit as: bgp peerings

- bgp peerings are established on loopback interfaces
  - improved resiliency
    - the peering stays up even if all the router's physical interfaces are down
- two loopbacks for each border router of as10
  - **`ifconfig lo:1 2.2.2.2 netmask 255.255.255.255 up`**
    - **`lo:1`** is an ip alias used for the peerings
    - the usual loopback address, **`lo`**, is still available



beware: using **`ifconfig lo:1 2.2.2.2/32`** sets up a **`/0`** netmask instead(!)

- a default route would unexpectedly be announced when loopback interfaces are redistributed in an igp

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transit as: bgp peerings

- be careful when configuring peerings on the loopbacks
  - bgp complains if the source address of OPEN messages from a neighbor does not match the neighbor's address configured in the peering (in this case, the loopback address)
  - bgp messages come out of a physical interface, whose address is different from the loopback's
  - need to force the source address of bgp messages
    - **`update-source`**
- cisco says:

  You only have to use the ***`update-source`*** *command when someone is peering to your loopback address*

transit as: bgp peerings

zebra bgp configuration file

```
router bgp 10
   network 10.0.0.0/8
   network 12.0.0.0/30
   neighbor 1.1.1.1 remote-as 10
   neighbor 1.1.1.1 update-source 2.2.2.2
   neighbor 1.1.1.1 description as10rt1(iBGP)
   neighbor 3.3.3.3 remote-as 10
   neighbor 3.3.3.3 update-source 2.2.2.2
   neighbor 3.3.3.3 description as10rt3(iBGP)
```

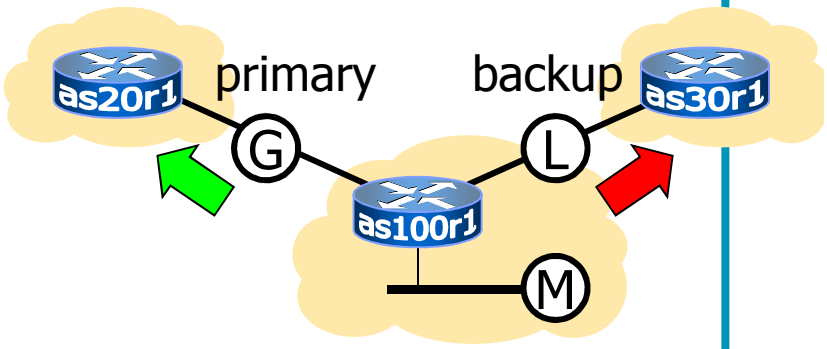
- note
  - **`update-source`** accepts an ip address or an interface name
  - zebra does not allow to set the **`update-source`** to an alias interface (e.g., **`lo:1`**)

transit as: some other flavouring

as100r1

```
as100r1:~# less /etc/zebra/bgpd.conf
hostname as100r1-bgpd
password zebra
!
ip prefix-list mineOut permit 100.0.0.0/8
!
route-map lowerPreference permit 10
   set local-preference 10
!
router bgp 100
   network 100.0.0.0/8
   neighbor 12.0.0.9 remote-as 20
   neighbor 12.0.0.9 description as20r1
   neighbor 12.0.0.9 prefix-list mineOut out
   neighbor 12.0.0.13 remote-as 30
   neighbor 12.0.0.13 description as30r1
   neighbor 12.0.0.13 prefix-list mineOut out
   neighbor 12.0.0.13 route-map lowerPreference in
```

/etc/zebra/bgpd.conf



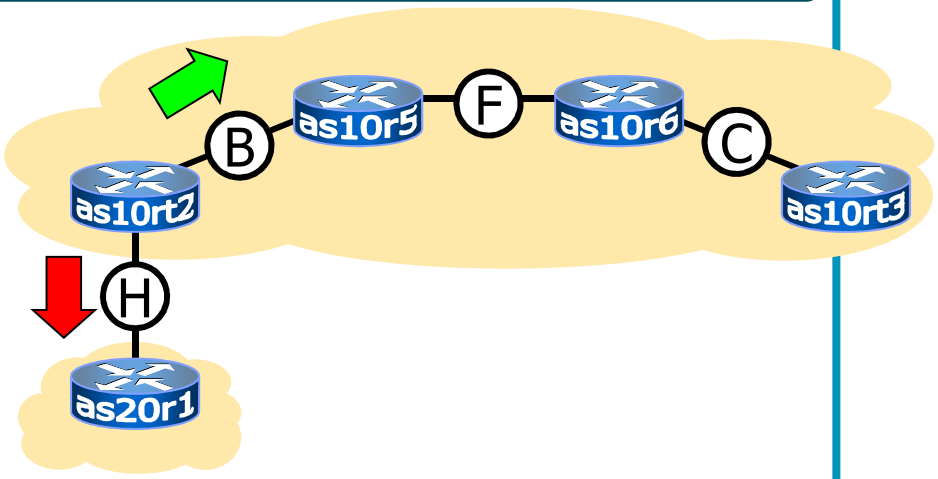
the customer prefers using link **G**

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transit as: some other flavouring

as10rt2

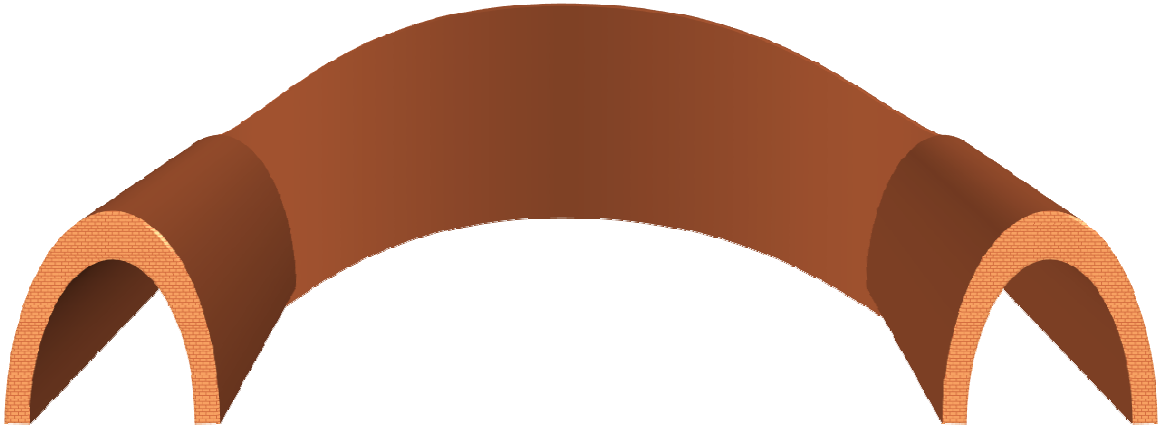
```
as10rt2:~# less /etc/zebra/bgpd.conf
hostname as10rt2-bgpd
password zebra
...
!
route-map dePref permit 10
   set local-preference 10
!
router bgp 10
   network 10.0.0.0/8
   network 12.0.0.0/30
   neighbor 1.1.1.1 remote-as 10
   neighbor 1.1.1.1 update-source lo
   neighbor 1.1.1.1 description as10rt1(iBGP)
   neighbor 3.3.3.3 remote-as 10
   neighbor 3.3.3.3 update-source lo
   neighbor 3.3.3.3 description as10rt3(iBGP)
   neighbor 12.0.0.2 remote-as 20
   neighbor 12.0.0.2 description as20r1(eBGP)
   neighbor 12.0.0.2 route-map dePref in
   neighbor 12.0.0.2 prefix-list noDefault in
/etc/zebra/bgpd.conf
```



as10rt2 prefers using the egress router **as10rt3**

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configuring a tunnel

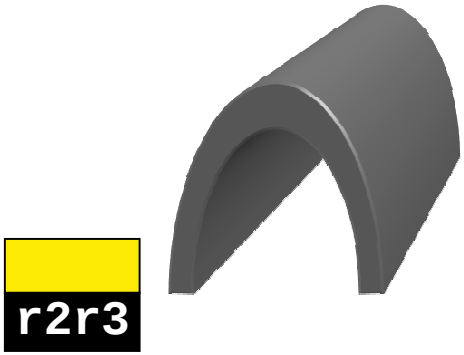


as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remote 1.1.1.1 local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

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configuring a tunnel



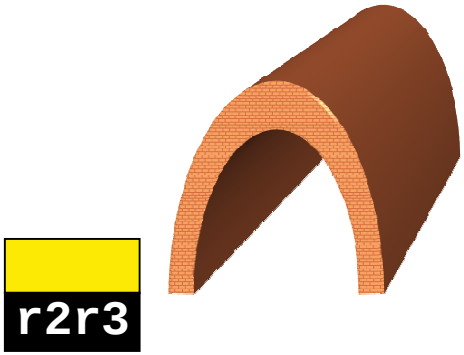
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig
ip tunnel                                  1.1.1.1 local 2.2.2.2 ttl 10
ip link s
ip addr a                                  3.0.0.5
ifconfig r2r1 up
```

endpoint name (appears as a virtual interface on the router

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configuring a tunnel



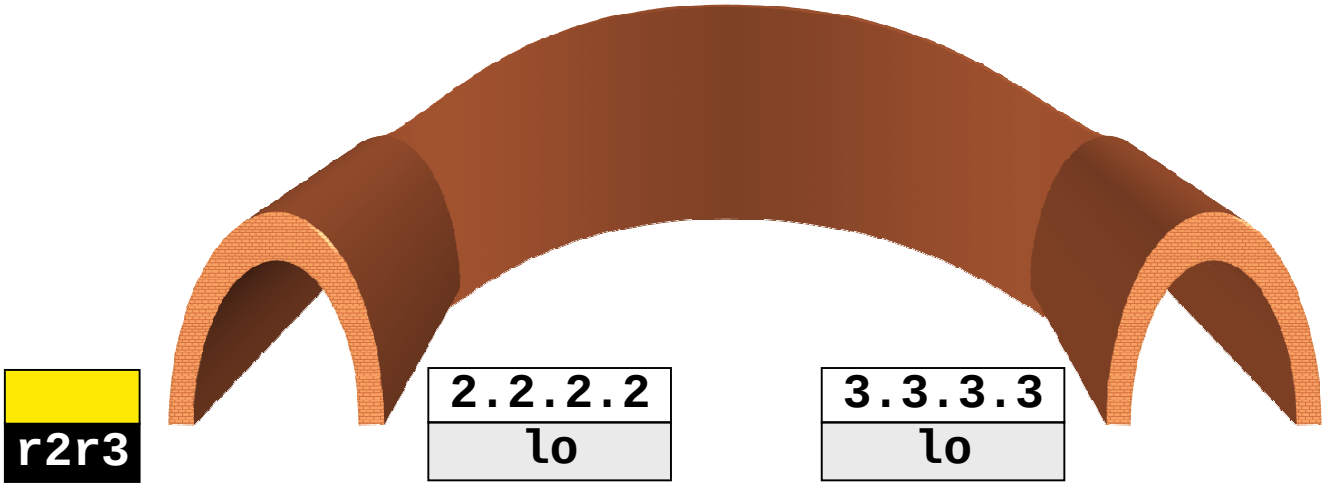
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.5 peer 13.0.0.6 ttl 10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remote 1.1.1.1 local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

encapsulation type
(IP in IP)

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configuring a tunnel



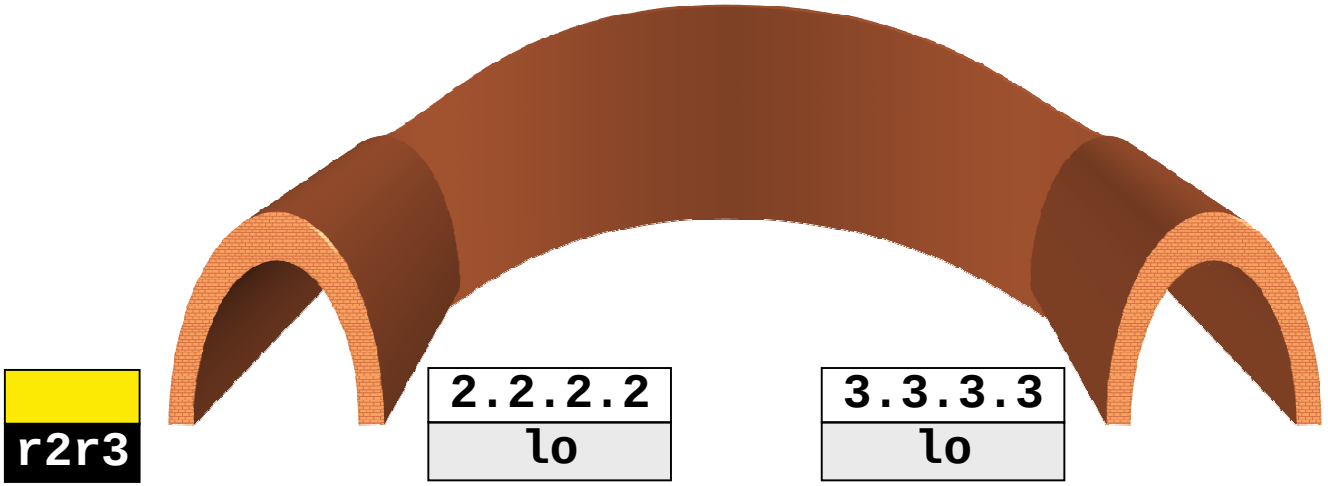
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9                              10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip r                             local 2.2.2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

tunnel endpoints

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configuring a tunnel



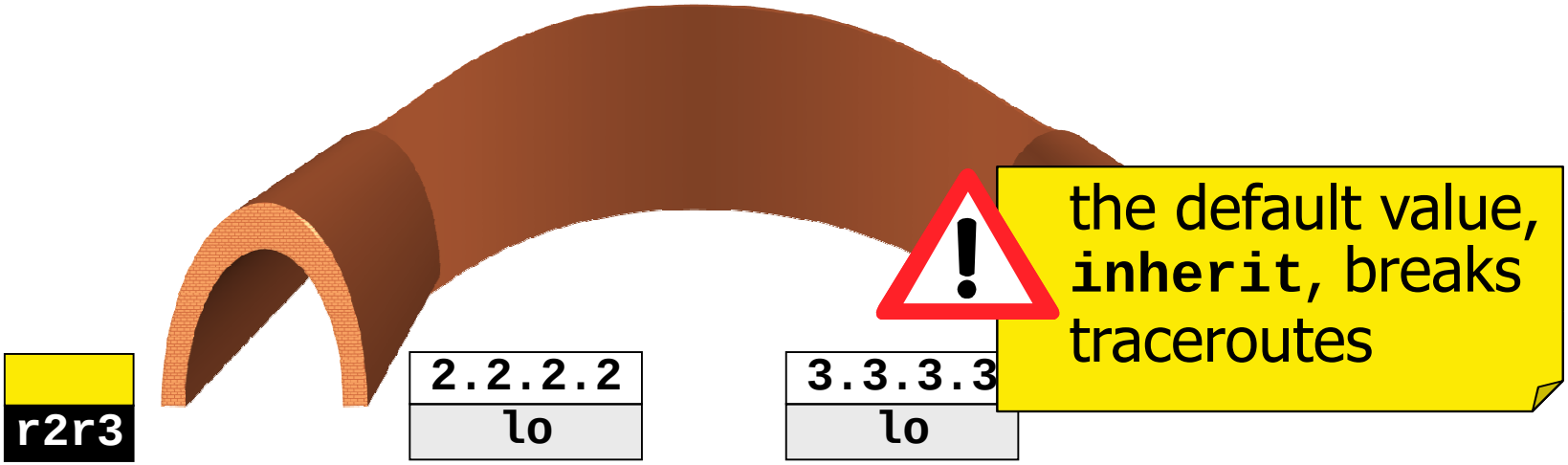
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode i
ip link set r2r1 multicas
ip addr add dev r2r1 13.0
ifconfig r2r1 up
```

ttl of the external envelope when a packet is encapsulated
(will expectedly travel through at most 4 hops, so any value >=4 should be fine)

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configuring a tunnel



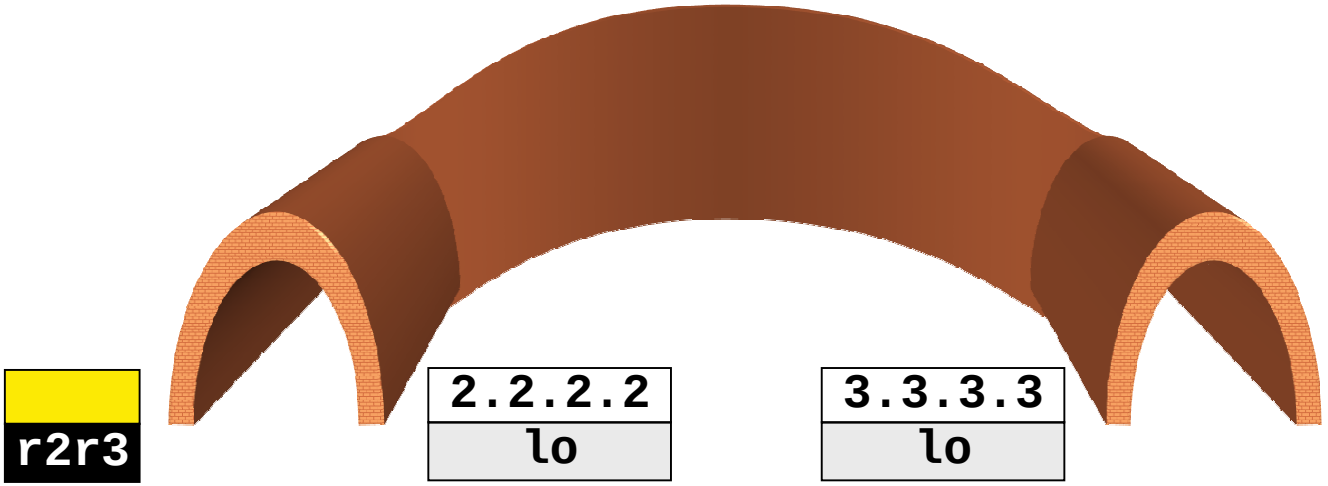
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode i
ip link set r2r1 multicas
ip addr add dev r2r1 13.0
ifconfig r2r1 up
```

ttl of the external envelope when a packet is encapsulated
(will expectedly travel through at most 4 hops, so any value >=4 should be fine)

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configuring a tunnel



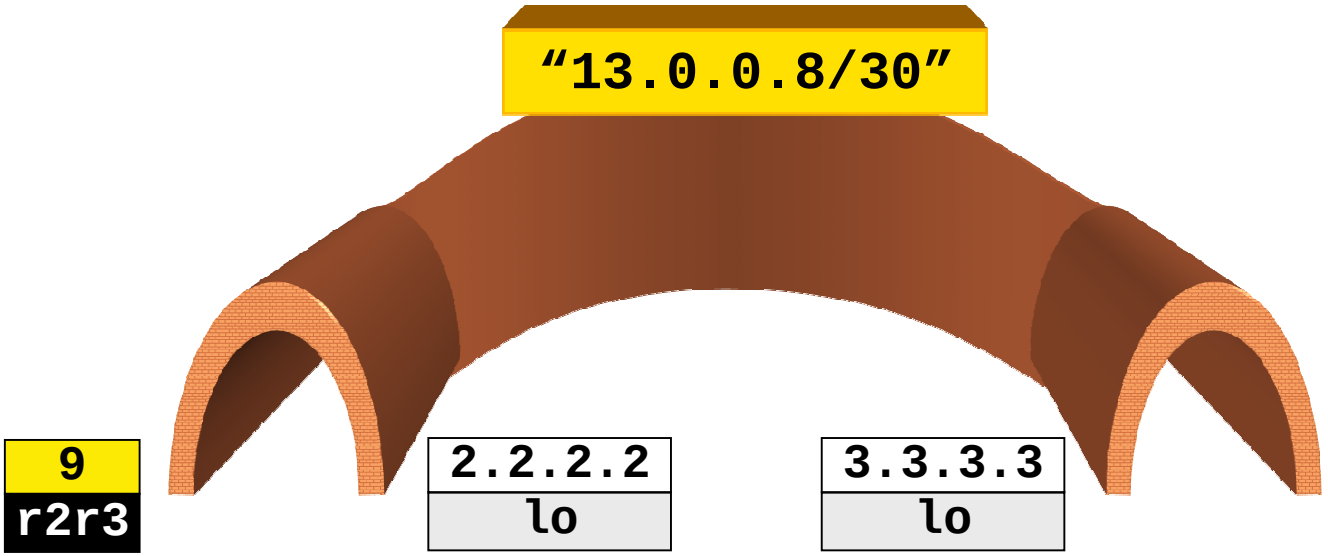
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 pee
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remot                    2.2 ttl 10
ip link set r2r1 multicast on
ip addr add dev r2r1 13.0.0.6 peer 13.0.0.5
ifconfig r2r1 up
```

rip uses multicast packets

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configuring a tunnel



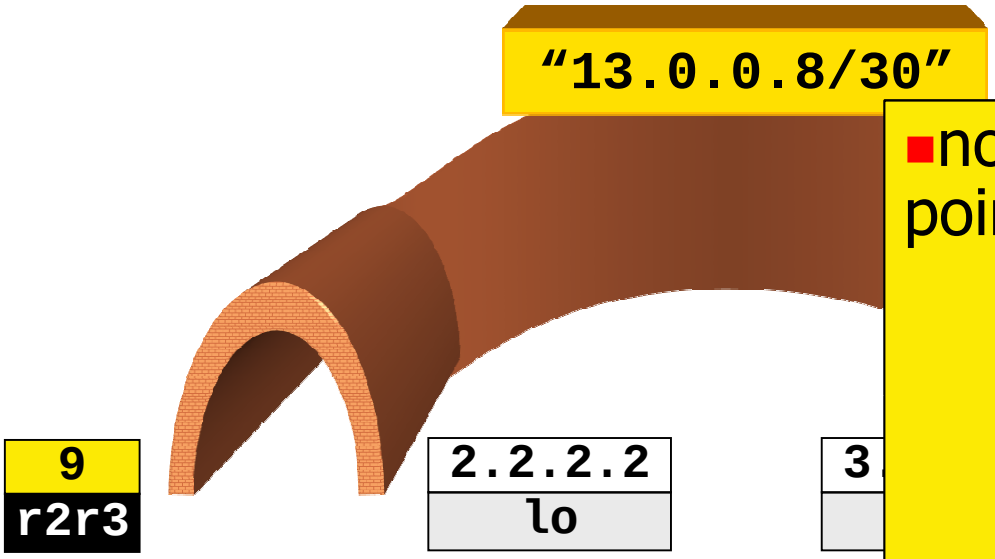
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
```

assign an ip address to the tunnel interface

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configuring a tunnel



- note: this is a pointopoint interface
  - speaking of “network” is senseless
  - we do it nevertheless to simplify the graphical layout

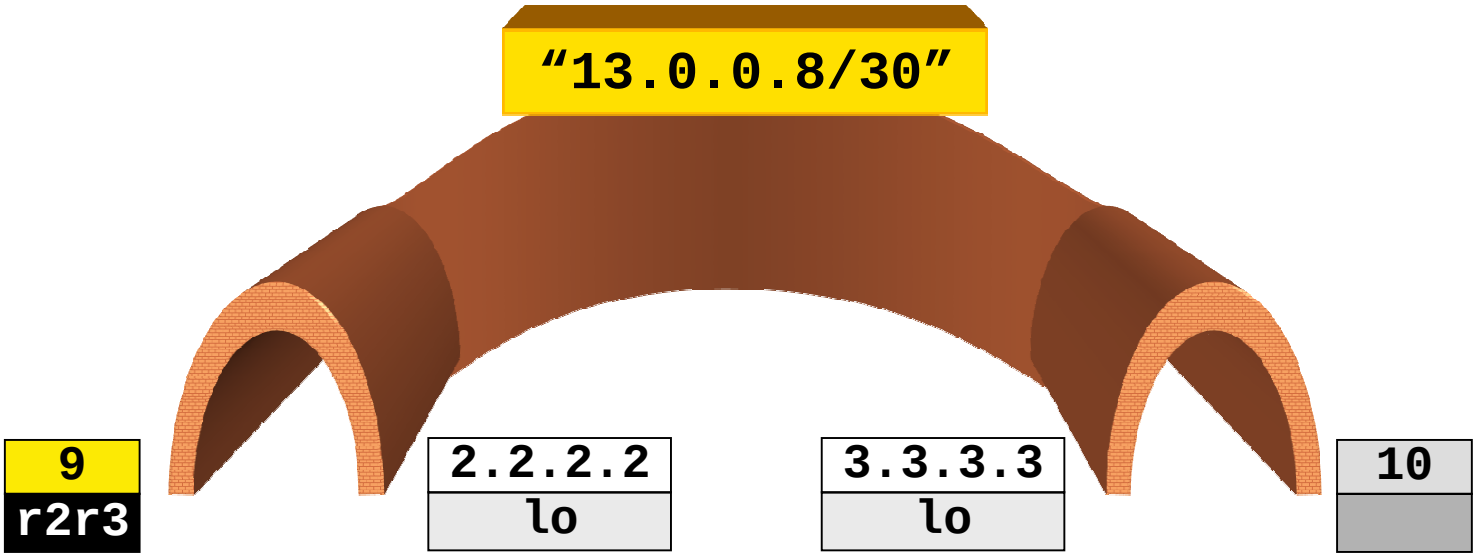
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mod          remote 1.1.1.1 local 2.2.2.2 ttl 10
ip lin
ip add                                r 13.0.0.5
ifconf
```

assign an ip address to the tunnel interface

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configuring a tunnel



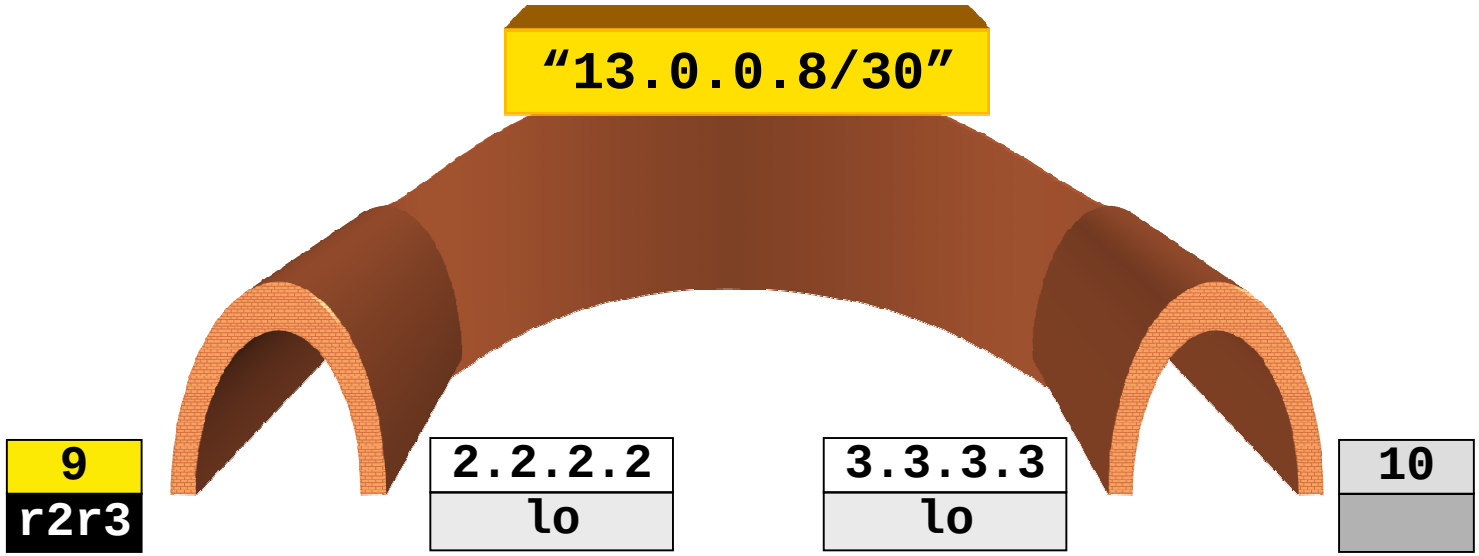
as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tun                .1.1.1 local 2.2.2.2 ttl 10
ip lin
ip add                .0.0.5
ifconf
```

for a pointopoint interface we can set the address of the other endpoint

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configuring a tunnel



as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
```

for a pointopoint interface we can set the address of the other endpoint

automatically inserts an entry in the routing table for **13.0.0.10**

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configuring a tunnel



note: failure to set the peer's address prevents rip from recognizing packets coming from the tunnel (possibly because it cannot match the sender's address with any of the local interface's subnets)

```
2007/10/30 11:27:25 RIP: RECV packet from
   13.0.0.10 port 520 on unknown
2007/10/30 11:27:25 RIP: packet comes from
   unknown interface
```

9
r2r3

interface name expected here

as10rt2 configuration

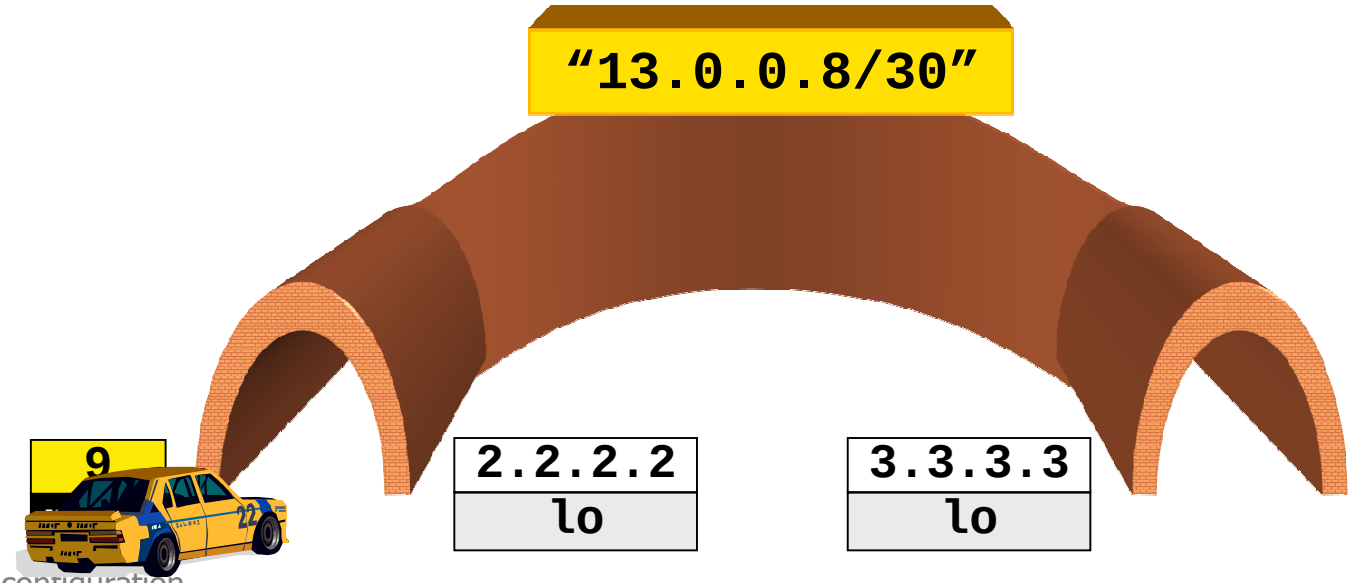
```
ip tunnel add r2r3 mode ipip remote 3.3.3.
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
```

for a pointopoint interface we can set the address of the other endpoint

automatically inserts an entry in the routing table for **13.0.0.10**

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configuring a tunnel



as10rt2 configuration

```
ip tunnel add r2r3 mode ipip remote 3.3.3.3 local 2.2.2.2 ttl 10
ip link set r2r3 multicast on
ip addr add dev r2r3 13.0.0.9 peer 13.0.0.10
ifconfig r2r3 up
ip tunnel add r2r1 mode ipip remote 1.1.1.1 local 2.2.2.2 ttl 10
ip link set r2r1 multi
ip addr add dev r2r1 13
ifconfig r2r1 up
```

switch the tunnel interface on

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```



tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNe
 distribute-list externalNe
 distribute-list internalNet
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

distribute also **lo:1**'s address (but not **lo**'s address because it is within the reserved **127.0.0.0/8**)

tunnels and routing

as10rt2 ripd configuration

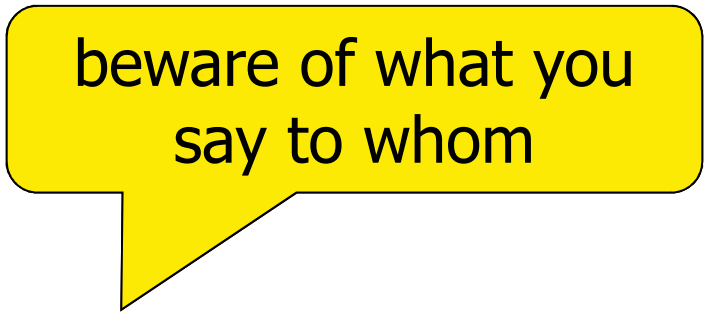
```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNe
 distribute-list externalNe
 distribute-list intern
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

propagate a statically configured default route *inside* the transit as (offers Internet connectivity to internal routers, if required)

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

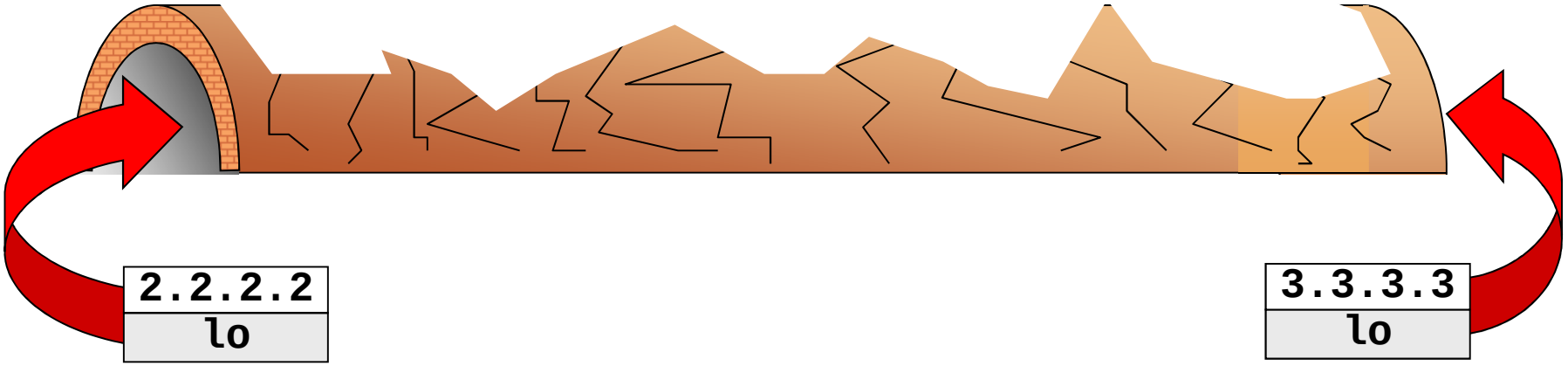


tunnels and routing



- we cannot announce the tunnel's endpoints inside the tunnel

tunnels and routing



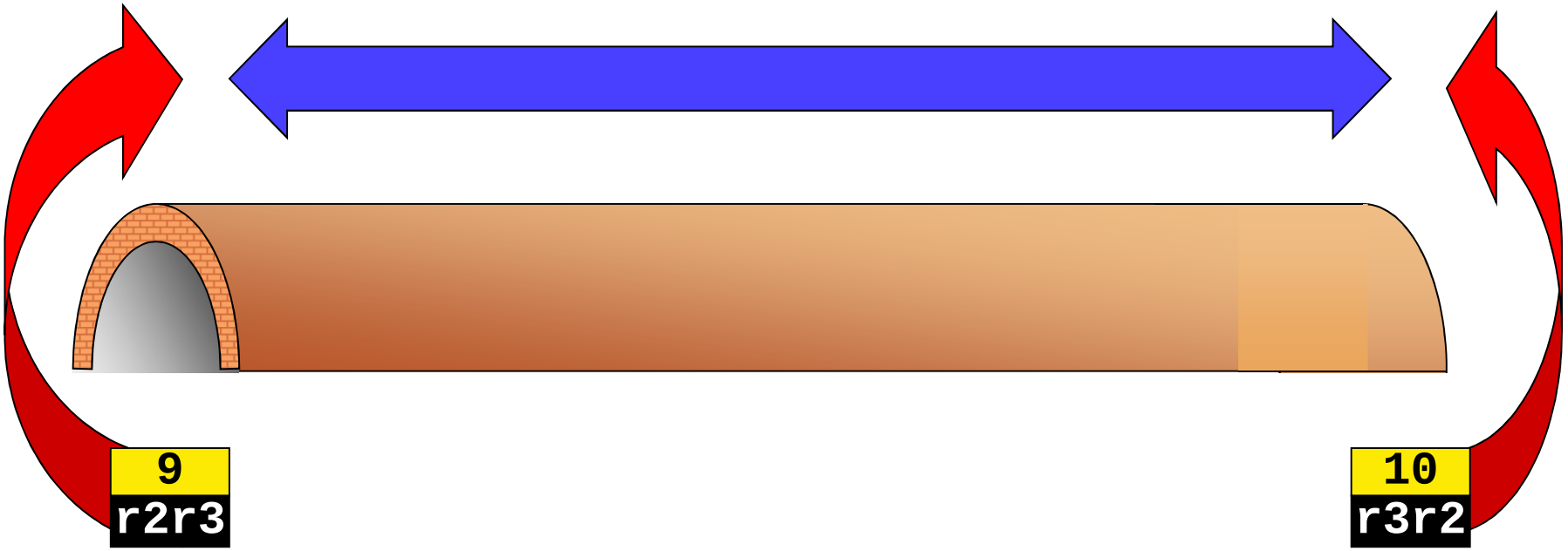
- we cannot announce the tunnel's endpoints inside the tunnel
  - that would tear down the tunnel!

tunnels and routing



- we shouldn’t announce the tunnel’s interfaces outside the tunnel

tunnels and routing



- we shouldn’t announce the tunnel’s interfaces outside the tunnel
  - traffic might flow outside the tunnel

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-l                                  ut r2r1
 distribute-l                                  ut r2r3
 distribute-l                                  ut eth1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

access-lists instruct rip about what to propagate

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-l                                    1
 distribute-l                                    3
 distribute-l                                    1
 route 0.0.0.0/0
!
access-list externalNetworks permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

ebgp next hops (in this case **as20r1**) are announced inside the tunnel

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-l                                  3
 distribute-l                                  1
 route 0.0.0.
!
access-list externa          works permit 12.0.0.0/30
access-list externalNetworks deny any
access-list internalNetworks deny 13.0.0.0/24
access-list internalNetworks deny 12.0.0.0/24
access-list internalNetworks permit any
```

ebgp next hops (in this case **as20r1**) are *not* announced outside the tunnel

tunnels and routing

as10rt2 ripd configuration

```
router rip
 redistribute connected
 network eth1
 network r2r3
 network r2r1
 distribute-list externalNetworks out r2r1
 distribute-list externalNetworks out r2r3
 distribute-list internalNetworks out eth1
 route 0.0.0.0/0
!
access-list externalNetworks per
access-list externalNetworks den
access-list internalNetworks den
access-list internalNetworks den
access-list internalNetworks per
```

note: the same routing behavior could be obtained using static routes

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tunnels and routing

- check the zebra routing table on **as10rt3**

as10rt3

```
Router> show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       B - BGP, > - selected route, * - FIB route

R>* 1.1.1.1/32 [120/4] via 11.0.0.9, eth1, 03:2
R>* 2.2.2.2/32 [120/4] via 11.0.0.9, eth1, 03:2
C>* 3.3.3.3/32 is directly connected, lo
R>* 10.0.0.0/30 [120/3] via 11.0.0.9, eth1, 03:
R>* 10.0.0.4/30 [120/2] via 11.0.0.9, eth1, 03:27:03
R>* 10.0.0.8/30 [120/2] via 11.0.0.9, eth1, 03:27:0
R>* 11.0.0.0/30 [120/3] via 11.0.0.9, eth1, 03:27:0
R>* 11.0.0.4/30 [120/3] via 11.0.0.9, eth1, 03:27:
C>* 11.0.0.8/30 is directly connected, eth1
R>* 12.0.0.0/30 [120/2] via 13.0.0.9, r3r2, 03:26:44
B   12.0.0.0/30 [200/0] via 2.2.2.2, recursive via 11.0.0.9, eth1, 03:26:58
C>* 12.0.0.4/30 is directly connected, eth0
B>* 12.0.0.8/30 [200/0] via 12.0.0.2, recursive via 13.0.0.9, r3r2, 03:26:38
B>* 12.0.0.12/30 [20/0] via 12.0.0.6, eth0, 03:26:24
C>* 13.0.0.2/32 is directly connected, r3r1
C>* 13.0.0.9/32 is directly connected, r3r2
B>* 20.0.0.0/8 [200/0] via 12.0.0.2, recursive via 13.0.0.9, r3r2, 03:26:38
B>* 30.0.0.0/8 [20/0] via 12.0.0.6, eth0, 03:26:24
B>* 100.0.0.0/8 [20/0] via 12.0.0.6, eth0, 03:26:24
C>* 127.0.0.0/8 is directly connected, lo
```

destinations routed through the tunnel

tunnels and routing

- **as10rt2** prefers the egress point **as10rt3**

as10rt2

```
as10rt2:~# traceroute -s 12.0.0.1 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1) from 12.0.0.1, 64 hops max, 40
byte packets
 1  13.0.0.10 (13.0.0.10)  3 ms  3 ms  2 ms
 2  12.0.0.6 (12.0.0.6)  2 ms  4 ms  5 ms
 3  100.0.0.1 (100.0.0.1)  2 ms  2 ms  2 ms
```

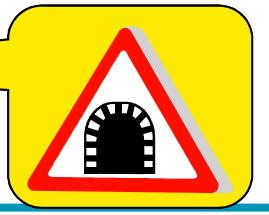
tunnels and routing

- **as10rt2** prefers the egress point **as10rt3**

did we already mention you should use a source address that is reachable from outside the transit as?

as10rt2

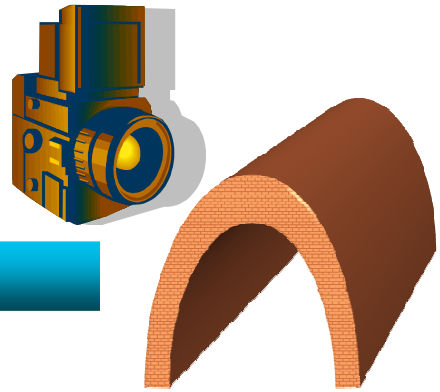
```
as10rt2:~# traceroute -s 12.0.0.1 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1) from 12.0.0.1, 64 hops max, 40
byte packets
 1  13.0.0.10 (13.0.0.10)  3 ms  3 ms  2 ms
 2  12.0.0.6 (12.0.0.6)  2 ms  4 ms  5 ms
 3  100.0.0.1 (100.0.0.1)  2 ms  2 ms  2 ms
```



- now **as10rt3** is directly reached via the tunnel

tunnels and routing

- a look outside the tunnel



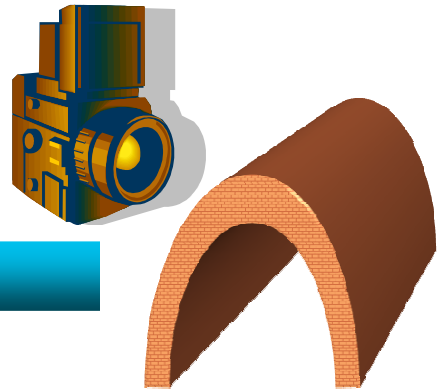
as10rt2

```
as10rt2:~# ip tunnel show r2r3
r2r3: ip/ip  remote 3.3.3.3  local 2.2.2.2  ttl 10
as10rt2:~# ip link show eth0
1: eth0: <BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether fe:fd:0c:00:00:01 brd ff:ff:ff:ff:ff:ff
as10rt2:~# ip link show r2r3
7: r2r3@NONE: <POINTOPOINT,MULTICAST,NOARP,UP> mtu 1480 qdisc noqueue
    link/ipip 2.2.2.2 peer 3.3.3.3
as10rt2:~#
```

the tunnel is active

tunnels and routing

- a look outside the tunnel

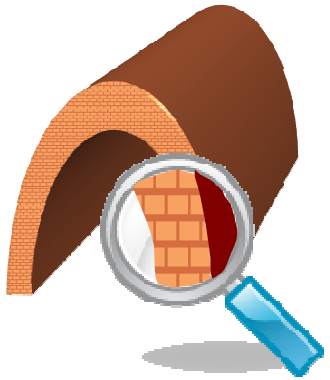


as10rt2

```
as10rt2:~# ip tunnel show r2r3
r2r3: ip/ip  remote 3.3.3.3  local 2.2.2.2  ttl 10
as10rt2:~# ip link show eth0
1: eth0: <BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether fe:fd:0c:00:00:01 brd ff:ff:ff:ff:ff:ff
as10rt2:~# ip link show r2r3
7: r2r3@NONE: <POINTOPOINT,MULTICAST,NOARP,UP> mtu 1480 qdisc noqueue
    link/ipip 2.2.2.2 peer 3.3.3.3
as10rt2:~#
```

the tunnel's **mtu** is 20 bytes smaller because of the additional ip header

tunnels and routing



- a look inside the tunnel

as10rt2

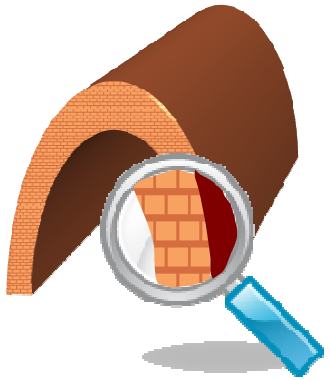
```
as10rt2:~# ping -I 12.0.0.1 100.0.0.1
PING 100.0.0.1 (100.0.0.1) from 12.0.0.1 : 56(84) bytes of data.
64 bytes from 100.0.0.1: icmp_seq=1 ttl=63 time=1.43 ms
64 bytes from 100.0.0.1: icmp_seq=3 ttl=63 time=1.41 ms
```

as10r5

```
as10r5:~# tcpdump -i eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol
decode
listening on eth1, link-type EN10MB (Ethernet), capture size 96 bytes
14:58:30.602023 IP 2.2.2.2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64: echo request seq 27 (ipip-proto-4)
14:58:31.612680 IP 2.2.2.2 > 3.3.3.3: IP 12.0.0.1 >
64: echo request seq 28 (ipip-proto-4)
```

packets are encapsulated

tunnels and routing



- a look inside the tunnel

as10rt2

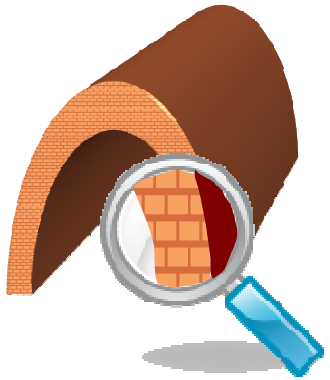
```
as10rt2:~# ping -I 12.0.0.1 100.0.0.1
PING 100.0.0.1 (100.0.0.1) from 12.0.0.1 : 56(84) bytes of data.
64 bytes from 100.0.0.1: icmp_seq=1 ttl=63 time=1.43 ms
64 bytes from 100.0.0.1: icmp_seq=3 ttl=63 time=1.41 ms
```

as10r5

```
as10r5:~# tcpdump -i eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol
decode
listening on eth1, link-type EN10MB (Ethernet), capture size 96 bytes
14:58:30.602023 IP 2.2.2.2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64: echo request seq 27 (ipip-proto-4)
14:58:31.612680 IP 2.2.2.2 > 3.3.3.3: IP 12.0.
64: echo request seq 28 (ipip-proto-4)
```

outer ip addresses correspond to the tunnel endpoints

tunnels and routing



- a look inside the tunnel

as10rt2

```
as10rt2:~# ping -I 12.0.0.1 100.0.0.1
PING 100.0.0.1 (100.0.0.1) from 12.0.0.1 : 56(84) bytes of data.
64 bytes from 100.0.0.1: icmp_seq=1 ttl=63 time=1.43 ms
64 bytes from 100.0.0.1: icmp_seq=3 ttl=63 time=1.41 ms
```

as10r5

```
as10r5:~# tcpdump -i eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol
decode
listening on eth1, link-type EN10MB (Ethernet), capture size 96 bytes
14:[illegible]2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64:[illegible]ipip-proto-4)
14:[illegible]2 > 3.3.3.3: IP 12.0.0.1 > 100.0.0.1: icmp
64:[illegible]ipip-proto-4)
```

inner ip addresses correspond to the real source and destination

transit as: routing tables

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
R(n) 0.0.0.0/0          11.0.0.10             2 11.0.0.10       02:48
R(n) 1.1.1.1/32         10.0.0.10             3 10.0.0.10       02:59
R(n) 2.2.2.2/32         10.0.0.5              3 10.0.0.5        02:49
R(n) 3.3.3.3/32         11.0.0.10             2 11.0.0.10       02:48
R(n) 10.0.0.0/30        10.0.0.5              2 10.0.0.5        02:49
C(i) 10.0.0.4/30        0.0.0.0               1 self
C(i) 10.0.0.8/30        0.0.0.0               1 self
R(n) 11.0.0.0/30        10.0.0.10             2 10.0.0.10       02:59
R(n) 11.0.0.4/30        10.0.0.5              2 10.0.0.5        02:49
C(i) 11.0.0.8/30        0.0.0.0               1 self
as10r6-ripd>
```

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transit as: routing tables

as10r6

```
as10r6-ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

     Network            Next Hop         Metric From              Time
R(n) 0.0.0.0/0          11.0.0.10                                 48
R(n) 1.1.1.1/32         10.0.0.10                                 59
R(n) 2.2.2.2/32         10.0.0.5                                  49
R(n) 3.3.3.3/32         11.0.0.10                                 48
R(n) 10.0.0.0/30        10.0.0.5                                  49
C(i) 10.0.0.4/30        0.0.0.0
C(i) 10.0.0.8/30        0.0.0.0
R(n) 11.0.0.0/30        10.0.0.10                                 59
R(n) 11.0.0.4/30        10.0.0.5                                  49
C(i) 11.0.0.8/30        0.0.0.0
as10r6-ripd>
```

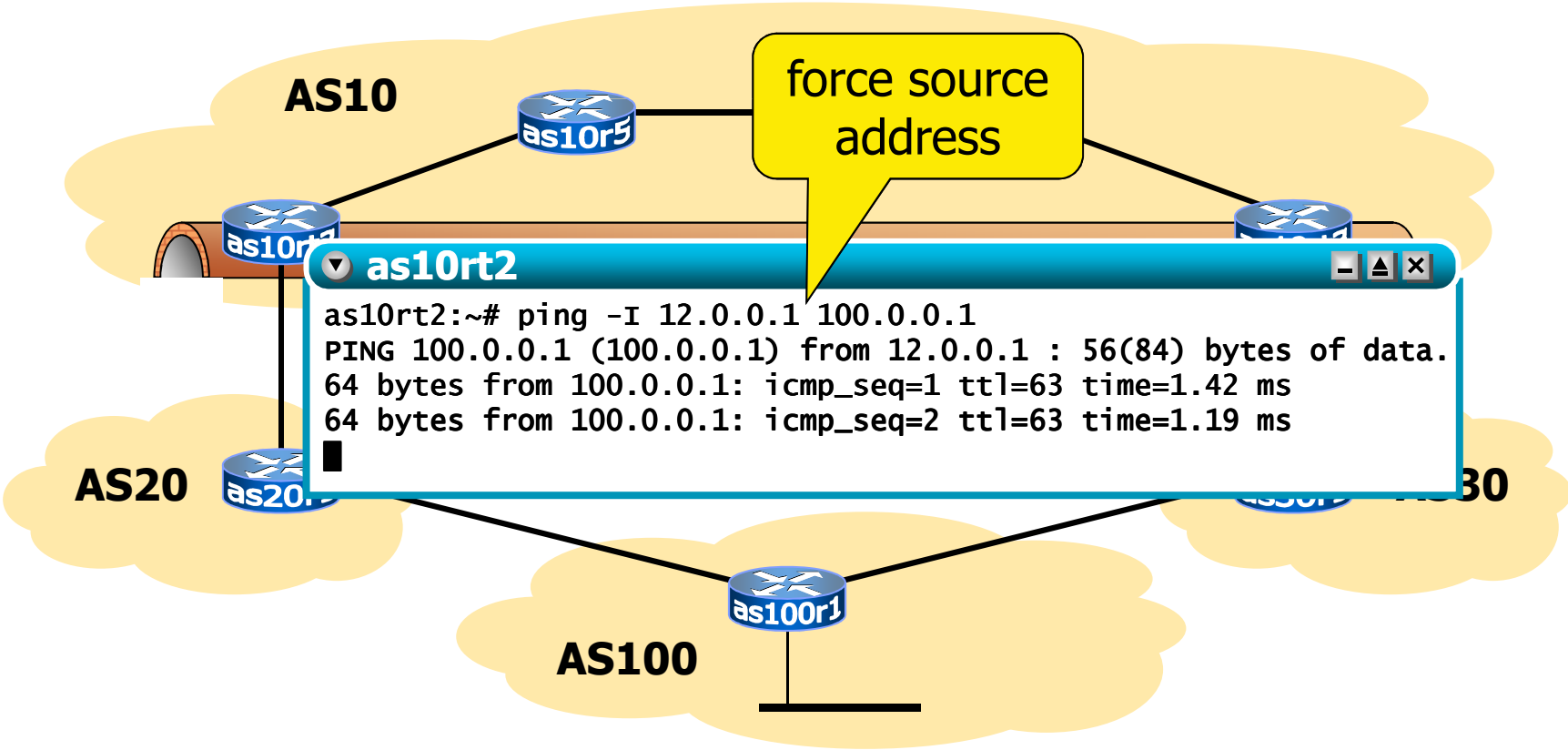
internal routers only know about internal destinations

(the default route is only there to offer Internet access, if required)

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transit as: asymmetric routing

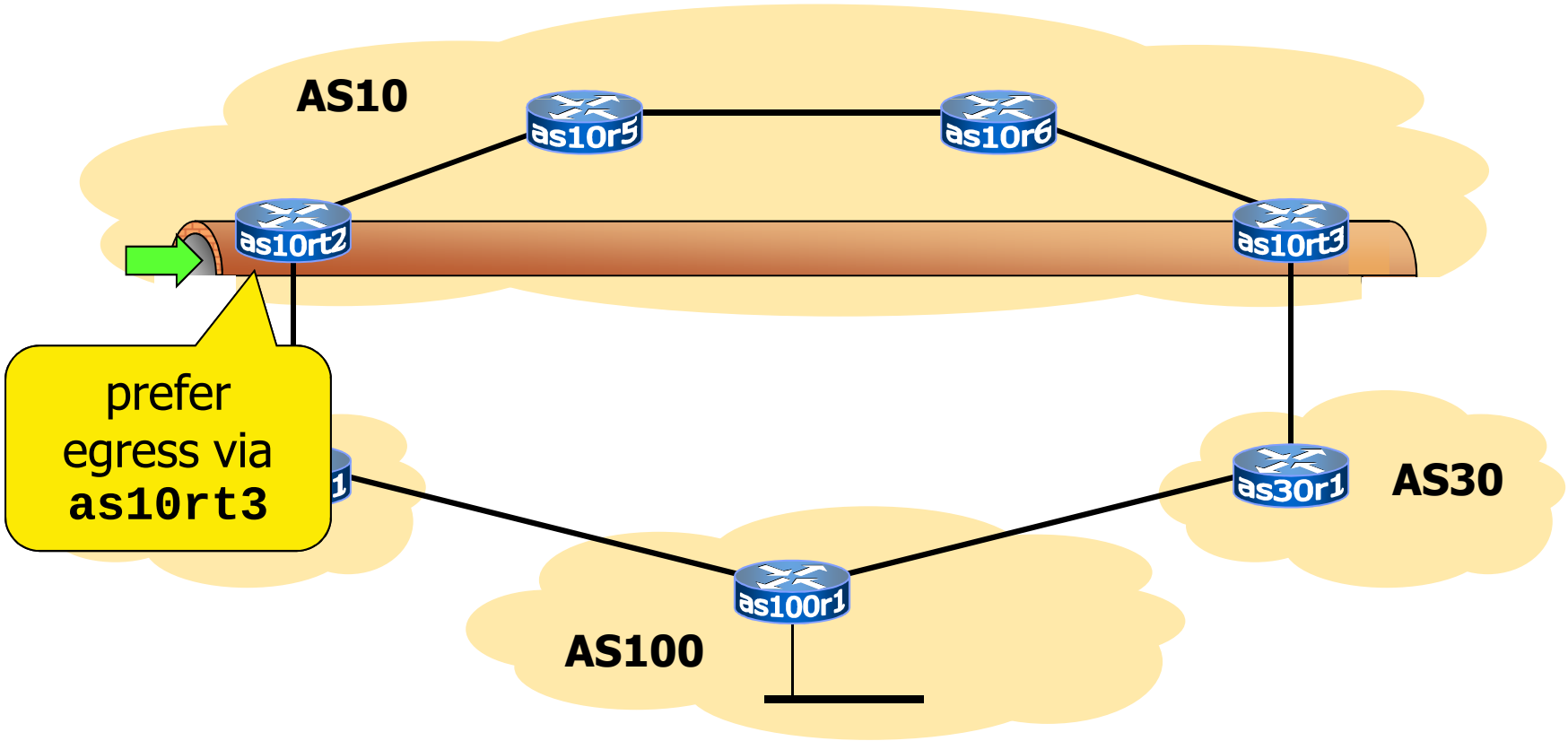
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

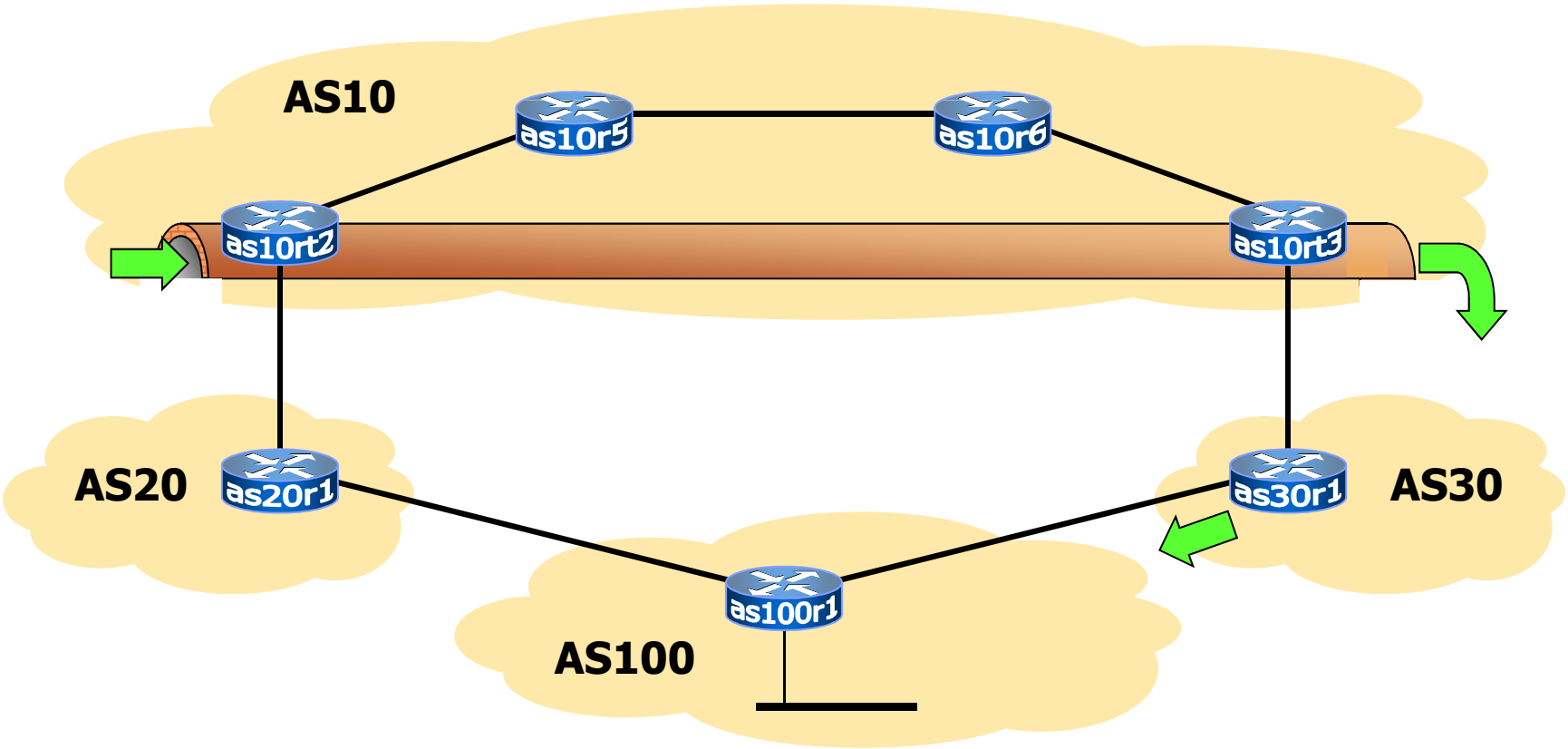
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

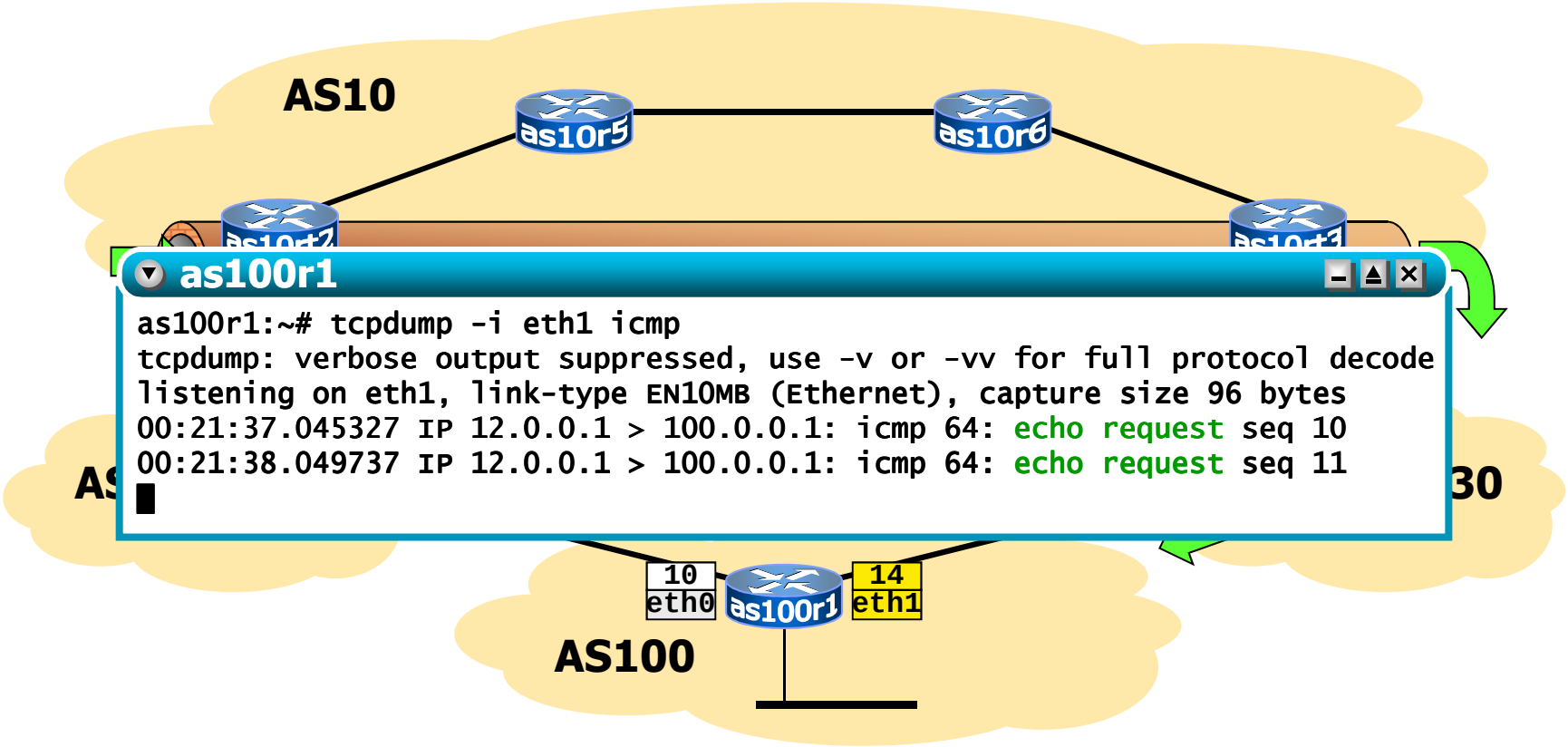
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

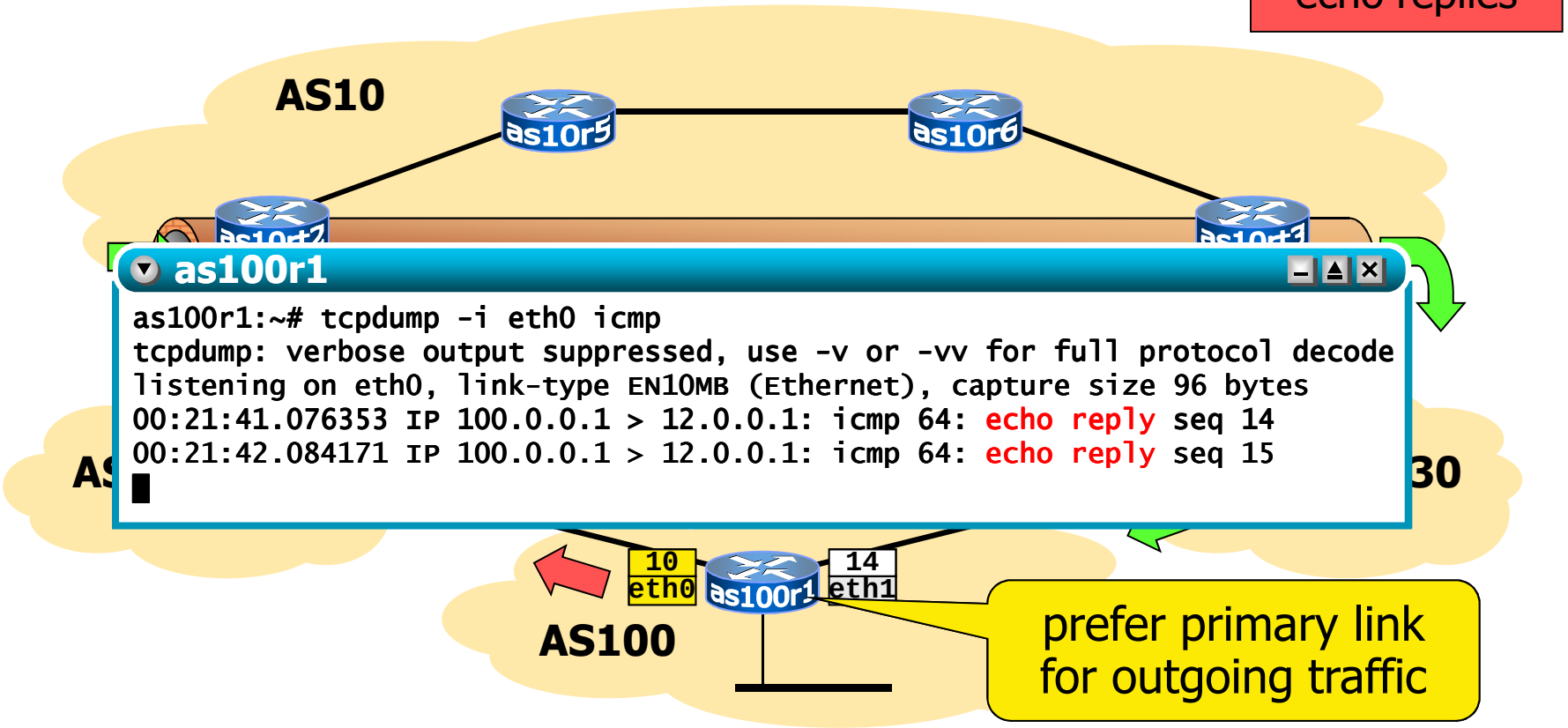
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

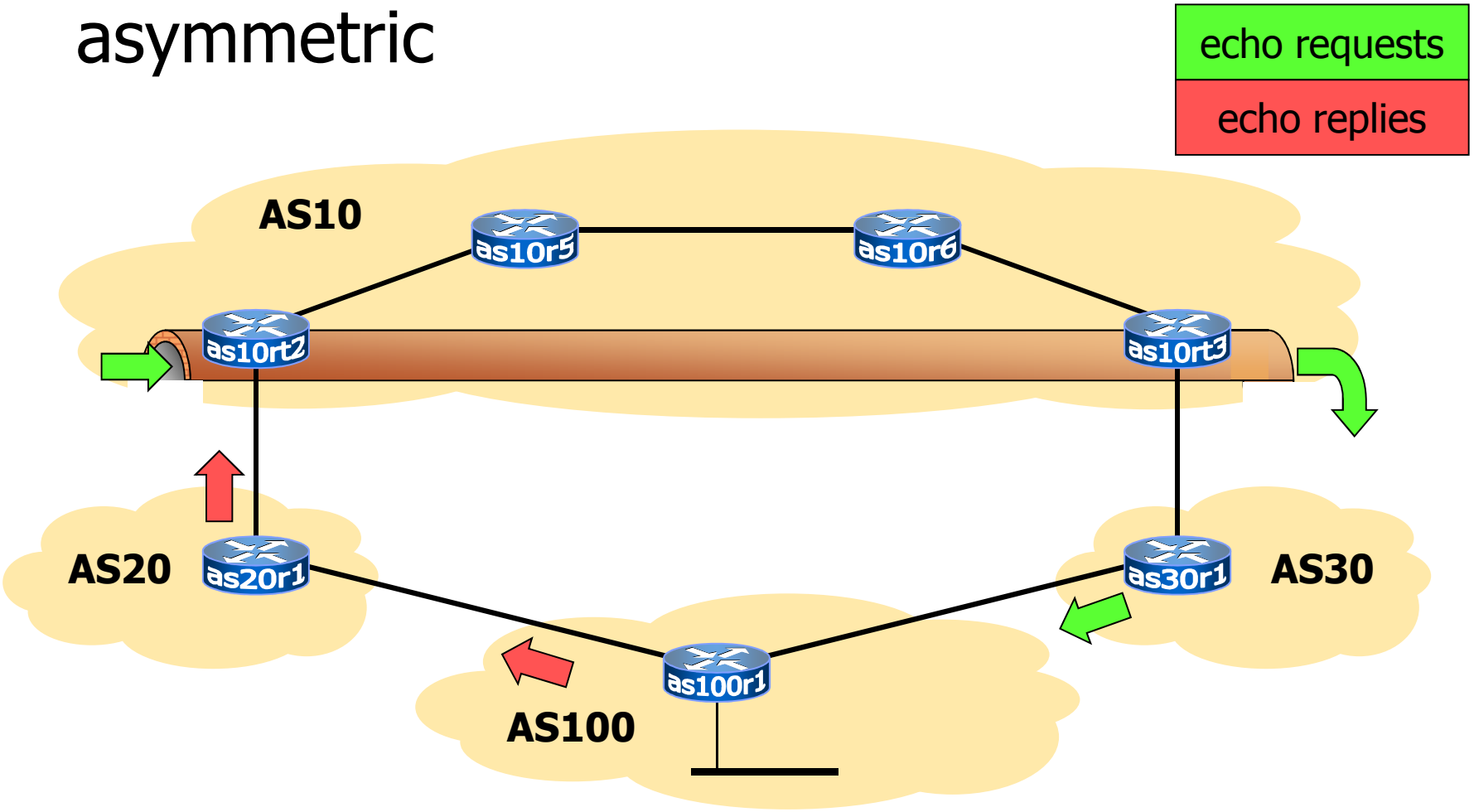
- bgp routing policies make routing asymmetric



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transit as: asymmetric routing

- bgp routing policies make routing asymmetric



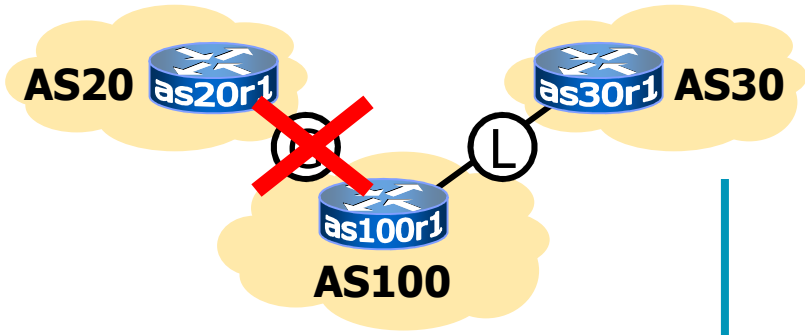
transit as: playing with the backup

- let's bring **as100**'s primary link down
  - expected result: traffic from **as20r1** to **as100r1** should traverse the transit as

as20r1

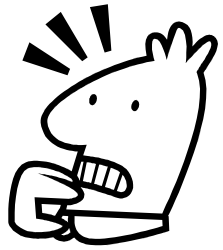
```
as20r1:~# telnet localhost bgpd
Trying 127.0.0.1...
Connected to as20r1.
Escape character is '^]'.

Hello, this is zebra (version 0.94).
...
as20r1-bgpd> enable
as20r1-bgpd# configure terminal
as20r1-bgpd(config)# router bgp 20
as20r1-bgpd(config-router)# neighbor 12.0.0.10 shutdown
```



transit as: playing with the backup

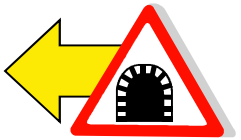
- wait for the routing to converge
  - fingers crossed...



- check the reachability of **100.0.0.0/8**

as20r1

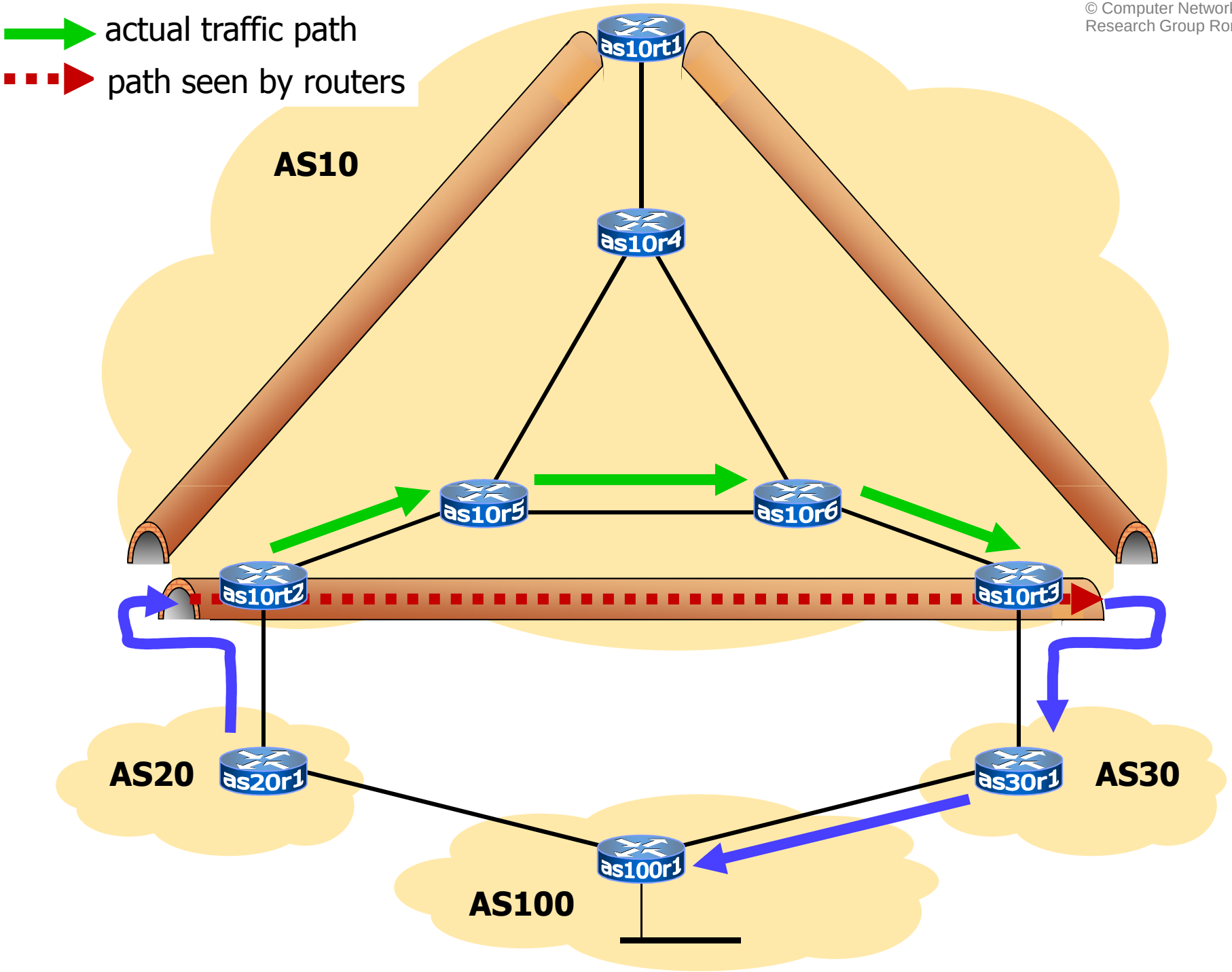
```
as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  2 ms  2 ms  1 ms
 2  13.0.0.10 (13.0.0.10)  2 ms  3 ms  2 ms
 3  12.0.0.6 (12.0.0.6)  3 ms  3 ms  3 ms
 4  100.0.0.1 (100.0.0.1)  3 ms  3 ms  3 ms
as20r1:~#
```





  - traffic is now traversing the transit as!

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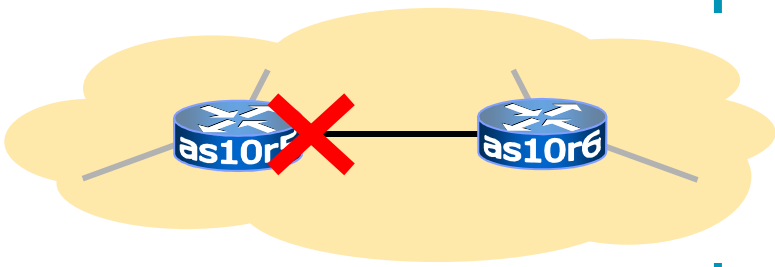
transit as: “rubbery” tunnels

- breaking an internal link does not tear the tunnels down
  - (as long as the transit as is not partitioned)

as10r5

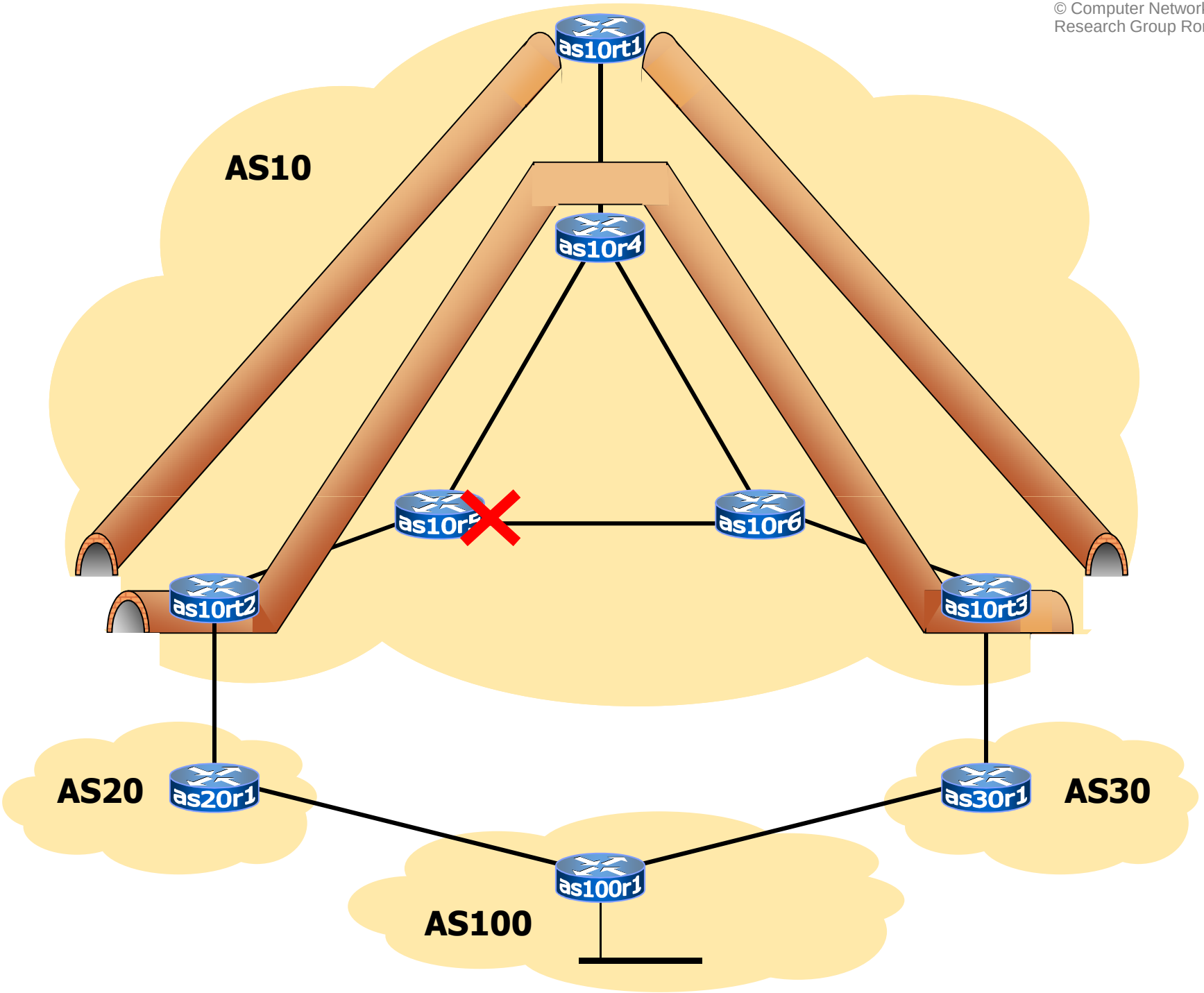
```
as10r5:~# ifconfig eth2 down
```






```
as10r5:~# traceroute 10.0.0.6
traceroute to 10.0.0.6 (10.0.0.6), 64 hops max, 40 byte packets
 1  10.0.0.1 (10.0.0.1)  1 ms  1 ms  1 ms
 2  10.0.0.6 (10.0.0.6)  1 ms  2 ms  2 ms
as10r5:~#
```

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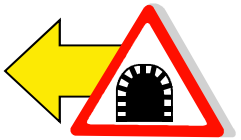


transit as: “rubbery” tunnels

- wait for the routing to converge
  - be **really** patient
- check the reachability of **`100.0.0.0/8`**

as20r1

```
as20r1:~# traceroute 100.0.0.1
traceroute to 100.0.0.1 (100.0.0.1), 64 hops max, 40 byte packets
 1  12.0.0.1 (12.0.0.1)  1 ms  1 ms  1 ms
 2  13.0.0.10 (13.0.0.10)  2 ms  4 ms  2 ms
 3  12.0.0.6 (12.0.0.6)  3 ms  3 ms  3 ms
 4  100.0.0.1 (100.0.0.1)  6 ms  3 ms  4 ms
as20r1:~#
```



  - traffic is still able to traverse the transit as

conclusions

- an overlay network is better
  - smaller routing tables on internal routers
  - less churn
  - predictable interplay between igp and egp
- sample implementation: tunnels
  - directed to the egress points
- observations
  - bgp peerings could be established on the tunnel interfaces
  - tunnels are as robust as the underlying igp